

# A manifest of docs/reports

One hundred and fourteen independent mathematical research packages,  
classified by subject

Vladimir Reshetnikov

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## 1 Scope and conventions

This document catalogues the one hundred and fourteen research packages stored under `docs/reports`. The collection originally held one hundred and four reports; after two consolidations it contributes one hundred and two maintained reports. The other twelve came through ProveIt’s incoming-reports procedure (`docs/incoming`) — six in batch 36, one each in batches 40, 41 and 44, two in batch 60 and one in batch 62 — whose archives remain in the repository’s history. Later batches also extended existing reports; their entries name the batch. The next paragraphs describe the historical collection before those two consolidations. On 29 September 2026 the two Lipparini Problem 6.2 reports were unified, and the finite-height transfinite-word report was integrated with its guarded-block companion. Both proof routes and all distinct extensions are retained; old directories contain redirects, not extra catalogue entries. The seven surreal-number reports that this catalogue used to carry were moved out to a repository of their own, with their history, and are no longer listed here. They arrived as one hundred and twenty-five ZIP archives, in seven deliveries, which have been unpacked and grouped into the subject hierarchy described in Section 2. The archives themselves are no longer kept in the repository.

One hundred and twenty-five archives yield one hundred and three reports for two different reasons. Twenty archives duplicated another: sixteen pairs, three three-way clusters and one four-way cluster. Two more were folded into a neighbour not because they repeated it but because they shared a subject — the three reports on term counts in the derivatives of power towers are now one article with a common framework and three parts. That consolidation is editorial, and the distinction matters: those three were put through the duplicate sweep and cleared it, and their three theorems are three different theorems. The seventh delivery of eighteen contained no duplicate at all, which is worth recording: a sweep that finds nothing is evidence about the collection, not a sweep that failed. Each pair has been merged into one report that proves the shared theorem once and keeps everything both originals built on top of it. Where the two reached a result by genuinely different routes, both proofs are kept and marked as alternatives, with a sentence on what each one buys; the merged shuffle report keeps three independent finite certifications of one theorem, and the merged report on surreal option graphs kept two different five-vertex counterexamples for the same reason before it moved out with the other surreal-number reports. A merged entry below names both of its source archives and says what was doubled.

Not every pair aimed at one problem is a duplicate. The two Gonshor reports both prove his product-birthday bound, but on domains that are incomparable in both directions, so neither theorem contains the other and they are catalogued separately; the same holds for reports answering different numbered problems of one paper, noted where it occurs.

Every package is self-contained and follows the same shape: a typeset article (`*.pdf`) with its complete L<sup>A</sup>T<sub>E</sub>X source, a `README.md` summarising the result, usually a `code/` directory with exact-arithmetic verification scripts, a `data/` or `results/` directory holding the recorded output of those scripts, and often `STATUS.md`, `sources.md` or `notes/` files auditing the sources consulted and the precise limits of the claim.

Two cautions apply uniformly and are therefore not repeated in the individual entries. First, these are AI-assisted research drafts: none has been independently refereed, and none has been checked in a proof assistant. Several continue a Lean or Rocq project elsewhere in ProveIt; that

confers no formal status on them, and their entries say which project they continue. Second, the descriptions below record *what each report sets out to settle and what it claims to prove*. Compiling this manifest involved reading each package’s statement of its problem and result; it did not involve checking any proof. Where a report itself flags a limitation — a prior disproof, an imported theorem, a correction to the source, or an exception left open — that is noted, because it changes what the report is.

The naming of the directories is editorial. Several distinct reports attack the same published problem, and some arrived under colliding archive names, so each directory has been given a name describing its own angle on the problem; the merged directories usually name their combined scope. The September 29 consolidations retain the base directory names to preserve existing links. The original archive names are recorded with every entry.

## 2 The directory tree

| Category                             | Subcategory                       | Reports    |
|--------------------------------------|-----------------------------------|------------|
| enumerative-combinatorics            | —                                 | 22         |
| ordinals-and-order-types             | ordinal-arithmetic                | 3          |
|                                      | wqo-powersets-and-statures        | 4          |
|                                      | friendly-order-types              | 3          |
|                                      | games-on-ordinals                 | 3          |
|                                      | transfinite-words                 | 1          |
|                                      | (top level)                       | 5          |
| generating-functions-and-asymptotics | oeis-sequence-asymptotics         | 9          |
|                                      | binary-product-coefficient-counts | 3          |
|                                      | (top level)                       | 5          |
| log-concavity-and-unimodality        | —                                 | 12         |
| hankel-determinants                  | catalan-and-ballot                | 4          |
|                                      | somos-and-elliptic                | 3          |
|                                      | growth-and-runs                   | 2          |
|                                      | (top level)                       | 1          |
| congruences-and-valuations           | supercongruences                  | 4          |
|                                      | iterated-series                   | 2          |
|                                      | (top level)                       | 3          |
| automata-and-formal-languages        | —                                 | 7          |
| tetration-and-digit-stabilization    | —                                 | 6          |
| graph-theory                         | —                                 | 3          |
| quaternionic-analysis                | —                                 | 1          |
| jacobian-conjecture                  | —                                 | 3          |
| galois-theory-and-radicals           | —                                 | 2          |
| hilbert-tenth-problem                | —                                 | 3          |
| <b>Total</b>                         |                                   | <b>114</b> |

More than a quarter of the collection consists of counterexamples and corrections rather than proofs. Ordinal Chomp, open-query games, compositional tree series, the secant numbers, the decimal-length bound for tetration stabilization, power-tower stabilization, iterated-power digit stabilization, cluster variables, chromatic coefficients of cycles, products of  $q$ -integers, domination roots, Hardinian array constants, the mixed-base counterexample, recursive exponents, Gregory sign thresholds and binary substitution discrepancy all refute, or replace with a proved statement, an assertion published or catalogued elsewhere. Several reports whose main outcome is a proof still correct the statement they prove: the Somos-8 certificates, the shifted Catalan Hankel polynomials and the Rueppel determinants record a sign error, a missing one-unit shift and a numbering ambiguity in their sources, and the independence-system thresholds report refines its target after finding that the original conjecture had already been disproved by someone else.

## 3 Ordinals and order types

### 3.1 Friendly order types

Three reports on Isa Vialard’s *friendly order type*  $f(P)$ , the rank of the tree of open-ended bad sequences in which every chosen point has an incomparable partner in the current residual. All three prove the same finite formula; they differ in what they build on top of it, and in how they

prove it — one by a forest-rank monovariant that survives moves deleting a whole upset, one through a lemma bounding the cut maxima. A fourth report proved the formula by the first report’s argument almost verbatim and has been merged into it.

### 1. Friendly Order Types Through Incomparability Components

ordinals-and-order-types/friendly-order-types/  
incomparability-components-and-infinite-tails/friendly\_order\_types.pdf  
from friendly\_order\_type\_research (1).zip

The same component formula, together with an exact friendly rank for connected infinite well-partial-orders: writing  $o(C) = \lambda + n$  with  $\lambda$  the limit part and  $T(C)$  the finite set of points with finite principal upsets. Includes an exhaustive finite verification and a proof audit.

### 2. Friendly Order Type Through Incomparability Components

ordinals-and-order-types/friendly-order-types/  
matroid-rank-and-ordinal-sums/article.pdf  
from friendly\_order\_type\_research (3).zip

Identifies  $|P| - c(P)$  with the graphic-matroid rank and the incidence-matrix rank of the incomparability graph. Adds a substitution formula, a finite Cartesian-product formula, and additivity over arbitrary well-ordered ordinal sums; evaluates every well-partial-order whose incomparability components are finite, and gives countable counterexamples delimiting stronger infinite extrapolations.

### 3. Friendly Order Types of Finite Posets and Ordinal Products

ordinals-and-order-types/friendly-order-types/  
components-forests-and-ordinal-products/friendly\_order\_types.pdf  
from friendly\_order\_types\_research.zip + friendly\_order\_type\_research.zip

Proves  $f(P) = |P| - c(\text{Inc}(P))$  for a finite poset  $P$ , where  $c$  counts connected components of the incomparability graph (isolated vertices included), and classifies the witnesses: every maximum friendly subset is  $P$  minus one minimal root per component, counted by  $\prod |\text{Min}(C_i)|$ . With Vialard’s width-transfer theorem this gives ordinal width  $\omega^{f(P)}$  for the Dershowitz–Manna multiset ordering. On the transfinite side it proves  $f(\alpha_1 \times \cdots \times \alpha_r \times B) = \Theta$ , the natural product, except that it is the predecessor of  $\Theta$  when every  $\alpha_i$  is a successor and  $B$  has a greatest element, and classifies arbitrary finite ordinal boxes. Also: spanning-forest certificates, exact monotonicity under added comparisons, a bivariate generating function, and explicit four-point factors with equal  $(o, h, w, f)$  whose products differ. *Merged; the Cartesian-product formula is proved twice, once by corner deletion and once by a chain-grid spanning-subgraph argument.*

## 3.2 Well-quasi-ordered powersets and statures

Four reports on the open questions in Section 6 of Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard, *Measuring well-quasi-ordered finitary powersets* (arXiv:2312.14587v2), and on the corresponding topological invariant.

### 4. Cofinal strata and ordinal absorption

ordinals-and-order-types/wqo-powersets-and-statures/  
cofinal-strata-of-finitary-powersets/article.pdf  
from cofinal\_strata\_ordinal\_powersets.zip; batch 42: ProveIt\_Ordinal\_Heights\_Matchings.zip;  
batch 36: ProveIt\_Nonuniform\_Powerset\_Heights.zip, ProveIt\_Persistent\_Frontier\_Heights.zip,  
persistent\_frontier\_heights.zip, ProveIt\_Ordinal\_Heights\_Research.zip

For  $S_\alpha(Q)$ , a finite poset  $Q$  with each point replaced by a chain  $\alpha = \omega^\rho$ , proves  $o(K(S_\alpha(Q))) = \sum_k \omega^{\rho \otimes k} c_k(Q)$  and  $h(K(S_\alpha(Q))) = \alpha \cdot \text{height}(M(Q))$ , where  $c_k$  counts size- $k$  maximal antichains not dominated by a larger one. The maximal-order-type theorem allows nonuniform pure well-partial-order fibers and is proved by a finite cofinal-stratification argument. Parts II and III, merged in batch 36 from four manuscripts that each prove it, supply the nonuniform height formula the original left open. For positive pure fibers  $\omega^{\rho q}$ ,  $h(K)$  is the largest ordinary ordinal sum, along a chain of maximal frontiers, of the largest fiber leaving the frontier at each step

plus the largest fiber of the last frontier; after Cantor-normal-form refinement the same formula covers nonzero limit fibers. For arbitrary ordinal fibers, finite and successor ones included, the height is given instead by a finite dynamic program over downsets with fallback weight 1, and the frontier compression is not valid there. Adds exact point ranks, NP membership of the threshold problem for the explicit pure-block encoding, and a separate part for well-partial-order fibers with  $h(K(P_q))$  a positive pure power. Part IV, added in batch 42, shows that for height-two skeletons with uniform fibers the height is  $\alpha(\nu_{\text{ur}} + 1)$ , with  $\nu_{\text{ur}}$  the maximum uniquely restricted matching number, so deciding the height is NP-complete already when every fiber is  $\omega$ .

### 5. Intrinsic ordinal approximations for finitary powersets

ordinals-and-order-types/wqo-powersets-and-statures/  
intrinsic-ordinal-approximations/article.pdf  
from intrinsic\_ordinal\_approximations.zip

Studies the intrinsic invariant  $a_o(A) = \sup\{o(B) + 1 : \text{Idl}(B) \hookrightarrow A\}$ . Proves the universal bound  $a_o(A) \leq h(P_f(A))$  with equality exactly when every finitely generated downset of  $A$  is an algebraic dcpo, gives the closed formula  $h(P_f(\prod_i \alpha_i)) = \sup(\bigotimes_i (1 + a_i) + 1)$ , and shows the universal identity fails on  $\lambda + F$ .

### 6. Ordinal Order Maps and Finitary Powersets of Ordinal–Finite Grids

ordinals-and-order-types/wqo-powersets-and-statures/  
ordinal-order-maps-and-grids/ordinal\_order\_maps.pdf  
from ordinal\_order\_maps\_research.zip

Exact maximal order types and heights for finite Hoare powersets of  $\beta \times Q$  with  $Q$  a finite poset: for  $\beta = \omega^{\gamma_0} + \dots + \omega^{\gamma_{m-1}}$ ,  $o(\text{Mon}(Q, \beta)) = \bigoplus_{s \text{ isotone}} \omega^{\bigoplus_q \gamma_{s(q)}}$ .

### 7. Sharp ordinal bounds for Hoare powerspaces

ordinals-and-order-types/wqo-powersets-and-statures/  
hoare-powerspace-statures/article.pdf  
from hoare\_statures\_research.zip

For all Noetherian spaces,  $1 + \|X\| \leq \|HX\| \leq 2^{\|X\|}$ , with witnesses attaining both endpoints at every ordinal and  $T_1$  upper witnesses; the older  $\omega^\alpha$  bound is attainable at infinite input stature  $\alpha$  exactly when  $\omega \cdot \alpha = \alpha$ ; the complete spectrum at  $\omega + n$  is  $\omega \cdot j(U) + (j(Q) - j(U))$ . Extends the known well-quasi-order results of Abriola et al. to arbitrary Noetherian spaces.

## 3.3 Transfinite words

One unified report on Harry Altman’s finite-alphabet sharpness question (arXiv:2409.03199v2, Example 3.15), combining three source manuscripts. The common height-two formula is stated once. The finite-height induction and the more detailed guarded-tail investigation are retained as distinct routes.

### 8. Finite-alphabet transfinite words: exact order types and guarded periodic blocks

ordinals-and-order-types/transfinite-words/order-types-below-omega-squared/article.pdf  
from omega2\_order\_types\_proposed\_solution.zip + ordinal\_words\_research.zip +  
finite\_alphabet\_transfinite\_words.zip

Claims  $o(V_k(P)) = \omega^{\omega^{N_k(P)-1}}$  for  $|P| \geq 2$  and each positive finite  $k$ , using canonical finite atom posets with  $N_{k+1} = |P| + J(I_k(P)) - 1$ . At height two this is  $\omega^{\omega^{n+j(P)-2}}$ ; empty and singleton alphabets are explicit exceptions. Retains the selected-tail-family classification, its universal-only exception, chain/antichain extremality, exact-length strata, and canonical linear-time token comparison. The guarded proof has both natural-product and elementary amplification routes; the finite-height proof uses protected separators. Two independent implementations and their separate audit records are preserved. Neither an all-height union formula nor a higher-cutoff selected-family classification is claimed.

### 3.4 Ordinal arithmetic

Three reports on ordinal and infinitary arithmetic, including a consolidation of two independent proposed solutions to Problem 6.2 of Paolo Lipparini, *Monotone infinitary operations on ordinals* (arXiv:2505.00424v2): define and study the least weakly monotone,  $e$ -special strictly increasing operation on countable sequences of ordinals.

#### 9. An Explicit Formula for the Minimal $e$ -Monotone Infinitary Ordinal Sum

[ordinals-and-order-types/ordinal-arithmetic/lipparini-minimal-infinitary-sum/article/explicit\\_ordinal\\_sum.pdf](#)

from `lipparini_problem_6_2_proposed_solution.zip` + `ordinal_operation_research.zip`

The same three-case formula for the least admissible operation is proved by two retained routes: profile ranks with arithmetic absorption and finite-product lower bounds, and corrected blocks with threshold interpolation and finite-part restoration. The latter avoids the product-rank lemma; both import Lipparini's characterization of the weaker  $S$ . Retains the multiset quotient, bounded-poset heights, finite-cap ranks with alternative proofs, equality on comparable inputs, cardinality preservation, and counterexamples to grouping and continuity. The correction is zero, a finite increment, or one *natural* summand  $\omega$ . Both implementations and their separate verification records are preserved.

#### 10. Jacobsthal Products All at Once

[ordinals-and-order-types/ordinal-arithmetic/jacobsthal-products-all-at-once/article.pdf](#)

from `Jacobsthal_Products_All_at_Once.zip`

Question 2.16 of Altman, *Intermediate arithmetic operations on ordinal numbers* (MLQ 63 (2017) 228–242), asks for an all-at-once order-theoretic model of finite and infinitary Jacobsthal products, rather than one built from parenthesized binary multiplication or a limit. Gives a constructive affirmative answer: a Cantor-normal-form comparison rule on actual tuples or finite-support functions, invariant under order isomorphisms and extending the componentwise order, whose definition never evaluates a Jacobsthal product. Adds a restricted maximality characterisation and a finite-tail reduction. The associativity identity and the binary normal-form formula are classical and are not claimed here; the construction is not claimed to be preserved by arbitrary order embeddings.

#### 11. Pairwise Coherence Does Not Guarantee Completion

[ordinals-and-order-types/ordinal-arithmetic/](#)

[pairwise-coherence-of-infinitary-products/ordinal\\_product\\_coherence.pdf](#)

from `ordinal_product_coherence.zip`

Answers the pairwise-coherence sufficiency question of Lipparini, *Noncommutative infinitary semigroups* (arXiv:2605.28413v1, Remark 7.1(b)), negatively. On the two-element group  $C_2$ , retaining every finite product and declaring every infinite all- $e$  product to be  $e$  gives a partial system satisfying full pairwise regrouping, even for arbitrary partitions, yet with no injective extension defining all finite and  $\omega$ -indexed products; two elements is minimal. Adds: for any monoid  $M$  the canonical partial domain  $P(M)$  is coherent, and for directly finite  $M$  it completes exactly when the unit group is trivial; and for every  $r$  a system on  $3r + 4$  elements coherent through nesting depth  $r$  but conflicting at depth  $r + 1$ , so no fixed finite depth of the coherence test characterises completability.

### 3.5 Games on ordinals

#### 12. An ordinal Chomp transition at 2

[ordinals-and-order-types/games-on-ordinals/](#)

[ordinal-chomp-transition-at-two/ordinal\\_chomp\\_counterexample.pdf](#)

from `ordinal_chomp_transition_counterexample.zip` + `ordinal_chomp_counterexample.zip`

For a numerical semigroup  $S$  ordered by  $x \leq_S y$  iff  $y - x \in S$ , and the ordinal extension ordered by ordinary ordinal subtraction (so  $S^2$  is the *lexicographic*, not coordinatewise, square), Chomp on  $S$

is a second-player win while Chomp on  $S^\sigma$  is a first-player win for every  $\sigma \geq 2$ , and  $\omega \cdot 4 + 4$  is the *least* winning opening. Hence  $\text{ch}(\langle 4, 5, 6 \rangle) = \text{ch}(\langle 4, 6, 9 \rangle) = 2$ , a negative answer to Question 4.6 of Rivero Herrera (arXiv:2504.07317v3) and a refutation of Conjecture 4.1 of v1, already for three small natural-number generators. The proof reduces the infinite game to a finite spectral statement via a finite-window recurrence and an exact finite-state certificate, then transfers it by an explicit response procedure. *Merged; the shared  $\langle 4, 6, 9 \rangle$  certificates were verified to hold the same 253 rows entry for entry, and the two reports' poisoned and unpoisoned conventions are reconciled by an explicit dictionary.*

### 13. Every ordinal as a point-separating game value

[ordinals-and-order-types/games-on-ordinals/  
point-separating-game-values/ordinal\\_separation.pdf](#)  
from ordinal\_separation\_research.zip

For every ordinal  $\lambda$ , orders  $\{0, 1\}^\lambda$  lexicographically and gives it the topology of all downward-closed sets, and the refinement generated by downward-closed sets minus finite sets. The point-separating game value is exactly  $\lambda$  for both: the first space is compact  $T_0$ , the second compact  $T_1$ , and for infinite  $\lambda$  neither is Hausdorff. Taking  $\lambda = \omega^2 + 1$  supplies a compact  $T_1$  witness of cardinality continuum for Problem 1.2 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3), with no CH, forcing or large-cardinal hypothesis. Hausdorff witnesses are not claimed. This is a different problem of that paper from the one the open-query games report answers.

### 14. Open-query games on ordinal spaces

[ordinals-and-order-types/games-on-ordinals/open-query-membership-games/article.pdf](#)  
from ordinal\_membership\_games.zip; batch 52: ProveIt\_Finite\_Output\_Open\_Query\_Games.zip;  
batch 56: ProveIt\_Filter\_Profiles\_Open\_Query\_Complexity.zip; batch 65:  
ProveIt\_Closure\_Word\_Duality.zip; batch 68: ProveIt\_Universal\_Word\_Profiles.zip

A negative answer to Problem 1.3 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3): two copies of  $[0, \omega]$  each have set-membership number 1, but their topological sum has value 2. The general theorem computes the invariant for every nonempty Hausdorff space of finite Cantor–Bendixson height  $h$  as  $\lceil \log_2(h + 1) \rceil$  when the last nonempty derivative is a singleton and  $\lceil \log_2(h + 2) \rceil$  otherwise. Part II, added in batch 52, replaces the membership bit by one of  $q$  output labels: the optimal depth is  $\lceil \log_2 L \rceil$  for the least number  $L$  of ordered layers with open suffixes; on Hausdorff spaces of finite height  $L \leq (q - 1)h + s$ , sharply, with  $s$  the number of labels on the last derivative; a topological sum of components of depth at most  $d$  has depth at most  $d + \lceil \log_2 q \rceil$ , sharply; and computing the optimal depth is polynomial for two labels but NP-complete for three, even on disjoint chains. At  $q = 2$  it reproduces Part I. Part III, added in batch 56, answers Part II's question of when that bound is attained, for spaces whose derived set is finite and nonempty: a coloring with  $q_c$  colors needs exactly  $q_c + \tau(G_c)$  layers,  $\tau$  the directed feedback vertex number of a color digraph; the bound is attained exactly under a bin-partition condition on the partition capacities of the neighborhood filters; and at three labels a sum of three one-point free-ultrafilter spaces has value 2 while a sum of three convergent sequences has value 3. Part IV, added in batch 65, drops the separation and finiteness hypotheses for a fixed coloring: a layer word is admissible exactly when it contains as subsequences all reduced words whose nested color closure is nonempty. It classifies the finite-depth behaviors, shows that the longest such word has the length of the least Cantor–Bendixson height of a Hausdorff representative, gives  $q_c + \tau(G_c)$  layers whenever the second derivative is empty, bounds finite-poset obstructions by  $mq(q - 1)^{m-1}$  points, and proves the product law  $\rho(X \times Y) = \rho(X) + \rho(Y) - 1$  for that longest length  $\rho$ . Part V, added in batch 68, computes the worst case over colorings at every finite height on locally compact or sequential (more generally rank-accessible) Hausdorff spaces:  $\max_c L = h(q - 1) + \min(q, t)$  with  $t$  the size of the last nonempty derivative, also when  $t$  is infinite, attained by one coloring constant off a countable set; it adds product and sum laws and an inverse theory of layer and query-depth spectra. Local data determining an arbitrary fixed coloring at height two, and a characterization of the spaces attaining that worst case (rank accessibility is sufficient, not necessary), remain open.

### 3.6 Order-theoretic invariants without a subcategory

Five reports whose objects sit outside every subcategory above: the minor order on finite graphs, two on sites of filters on  $\omega$ , congruences of finite lattices, and a translation-invariant ideal.

#### 15. Ordinal complexity of graph minors

[ordinals-and-order-types/graph-minor-ordinals/graph\\_minor\\_ordinals.pdf](#)  
from [graph\\_minor\\_ordinals.zip](#)

For finite simple graphs of maximum degree at most two under minor containment, computes maximal order type, ordinal antichain-tree width and descending-tree height:  $o(D) = w(D) = \omega^{\omega \cdot 2}$  and  $h(D) = \omega$ , with exact values  $\omega^{b+k}$  and  $\omega^{b+k-1}$  for the subfamilies bounded by path size  $b$  and cyclic-component count  $k$ . The general theorem is that a wqo  $C$  of connected graphs with  $o(C) = \alpha < \varepsilon_0$  has finite disjoint-union closure of type  $\omega^\alpha$ , proved by a component-splitting argument, since minor containment is not multiset embedding. A corollary is that continuity fails for increasing unions of minor-closed classes. The report is explicit that it does not compute the maximal order type of the minor order on all finite graphs and settles no named conjecture.

#### 16. A two-valued obstruction to gluing Fréchet powers

[ordinals-and-order-types/frechet-two-valued-obstruction/article.pdf](#)  
from [frechet\\_two\\_valued\\_obstruction.zip](#)

For the presheaf  $F \mapsto A^\omega / \sim_F$  on proper filters on  $\omega$  containing the cofinite filter, with the dense Grothendieck topology, proves in ZF that it is a sheaf if and only if  $|A| \leq 1$ . The counterexample is an explicit matching family of constant sections using two values, on a covering sieve with countably many generators each of countable base; the distinction between the number of sieve generators and the size of the full sieve is essential to it. The question is Massas, *A Semi-Constructive Approach to the Hyperreal Line* (2023), p. 517 after Fact 5.10, repeated in the 2024 dissertation *Duality and Infinity*, p. 249.

#### 17. Bounded-character Fréchet filter sites

[ordinals-and-order-types/frechet-bounded-character-sites/article.pdf](#)  
from [filter\\_complexity\\_research.zip](#), [Frechet\\_Cardinal\\_Boundaries.zip](#)

Continues the previous entry's non-claim about other filter sites. For the site  $P_\kappa$  of free filters on  $\omega$  generated by fewer than  $\kappa$  sets modulo the cofinite filter, represents the Booleanization  $C_\kappa$  as a regular-open algebra on the free ultrafilters and proves that  $P(\omega)/\text{Fin}$  is order dense in it exactly for  $\kappa \leq \mathfrak{p}$ ; above  $\mathfrak{p}$  no complete Boolean homomorphism preserving the modulo-finite events exists in either direction. The atoms are the ultrafilters of character below  $\kappa$ , so  $C_\kappa$  is atomless exactly for  $\kappa \leq \mathfrak{u}$ ;  $\mathfrak{h}$  is the first size of a family of events that cannot be decided jointly on a dense set of regions. Computes the associated sheaf by Boolean mixtures, with full Boolean-valued transfer, countable saturation of ultrafilter specializations, and sizes  $\mathfrak{c}$ ,  $2^\mathfrak{c}$ ,  $2^{2^\mathfrak{c}}$  for the raw power and the two sheafifications. A three-stage example uses Millán's forcing model as an imported input. Merged from two manuscripts whose indexings differ by a successor shift; the second one's closure-based results are stated at successor stages only.

#### 18. A sharp cover bound for congruences of finite lattices

[ordinals-and-order-types/lattice-congruence-cover-bound/sharp\\_cover\\_bound.pdf](#)  
from [lattice\\_cover\\_bound\\_research.zip](#)

If a finite  $n$ -element lattice has an element with at least  $k \geq 3$  upper covers then  $|\text{Con}(L)| \leq 2^{n-k-1}$ , with equality exactly for the glued sums  $C_s \oplus M_k \oplus C_t$ . This replaces the multicolour Ramsey threshold  $R_{k-1}(k)$  of the source's Lemma 4.4 (arXiv:2603.11454) by  $k$  itself. Adds a sharp nonextremal gap, a fan-label ideal refinement, and a bound on the size of Czédli's skeleton in terms of congruence density. All congruence machinery, including Freese's maximum, is proved from scratch. The target is an informal suggestion in the source, not a numbered conjecture.

#### 19. A non-Baire translation-invariant ideal

[ordinals-and-order-types/non-baire-translation-invariant-ideal/article.pdf](#)

from non\_baire\_ideal\_research.zip

An affirmative construction for Question 1 of Filipów and Tryba, *Densities for sets of natural numbers vanishing on a given family* (J. Number Theory 211 (2020) 371–382; arXiv:2309.00982, p. 7): a translation-invariant ideal that is not Baire and carries a perfect translation-almost-disjoint family.

## 4 Automata and formal languages

Five reports on state complexity, synchronization and language hierarchies. The first two are each a merge: the shuffle bound arrived four times and the DFAO reversal bound twice.

### 20. The six-state shuffle bound

automata-and-formal-languages/shuffle-six-state-bound/shuffle\_six\_state.pdf

from shuffle\_six\_state\_research (1).zip + shuffle\_six\_state\_research.zip +

Shuffle\_Six\_State\_Theorem.zip + six\_state\_shuffle\_research.zip

For complete DFAs with  $m, n \geq 2$  and  $\min(m, n) \leq 6$ , the worst-case deterministic state complexity of the shuffle is  $F(m, n) = 2^{mn-1} + 2^{(m-1)(n-1)}(2^{m-1} - 1)(2^{n-1} - 1)$ , over a finite alphabet allowed to depend on  $m$  and  $n$ ; the unrestricted conjecture stays open. The route is a reachability theorem in the universal shuffle automaton — a subset is reachable exactly when it meets the initial row and the initial column — proved by induction on  $(m + n, |S|)$  down to a finite certificate obligation. *Merged from four reports. Two shipped the same 10,642 witnesses; a third supplies an independent database of 10,619 witnesses sharing no witness with them, so the finite step is certified twice over; a fourth adds a fifth analytic reduction that cuts the obligation to 453 certificates and replaces the solver search with a deterministic enumeration. All three routes are kept.*

### 21. A coloring obstruction to maximal binary DFAO reversal

automata-and-formal-languages/dfa-reversal-coloring-obstruction/article.pdf

from dfa-reversal\_research.zip + binary\_dfa-reversal\_research.zip; batch 39:

ProveIt\_Three\_Output\_Reversal.zip; batches 40–41: ProveIt\_DFAO\_Eventual\_Optimality.zip,

ProveIt\_Binary\_DFAO\_Reversal.zip, ProveIt\_Binary\_Reversal\_Exactness.zip; batch 43:

ProveIt\_Four\_Output\_Reversal.zip, ProveIt\_Four\_Output\_Reversal (1).zip; batch 67:

ProveIt\_Near\_Linear\_DFAO\_Reversal.zip, ProveIt\_DFAO\_Spectral\_Cancellation.zip

For transformations  $a, b$  on an  $n$ -element set and an output map to  $k$  labels, proves  $|\tau\langle a, b \rangle| \leq k^n - k! + g(k) < k^n$  for  $3 \leq k \leq n$ , with  $g(k)$  the largest order of a permutation of  $k$  symbols — the binary-alphabet nonattainment question of Davies, arXiv:1705.07150v2, Section 5. The mechanism is a collision-orbit graph, a rightmost singular factorization, and a cyclic relabeling bound. Adds an abelian-permutation extension, a structural universal bound matching the published lower construction on all 372 pairs with  $7 \leq n \leq 30$ , and polynomial-time missing-coloring certificates. *Merged; the two originals ran the same three-case argument, and both a qualitative short proof and the counting proof are kept.* Part II, added in batch 39, settles three outputs exactly:  $R_2(n, 3) = 3^n - 3(2^A + 2^B - 2) + B$  for every  $n \geq 7$ , with  $(A, B)$  the closest coprime split of  $n$ , together with rigidity of the extremizers; the formula matches all twenty-four  $k = 3$  rows of the finite table. Part III, merged in batches 40–41 from three manuscripts, settles every  $k \geq 4$  past an explicit threshold:  $R_2(n, k) = k^n - \text{Bip}_k(a_n, b_n) + a_n b_n$  once  $n \geq N(k) \leq 3k^4$ , and for all  $n \geq \max(7, k + 1)$  when  $k = 4, \dots, 7$  (computer-assisted below  $N(k)$ ), with the lower construction imported from Davies; a non-effective eventual route for every  $k \geq 3$ , 919 finite certificates and a second proof of the  $k = 3$  formula are kept. Explicit exactness for  $k \geq 8$  below  $\min(N(k), T(k))$ , with  $T(k)$  the threshold of Part V, stays open. Part IV, merged in batch 43 from two manuscripts on  $k = 4$ , re-derives that case with an analytic threshold  $n \geq 26$  (against 212), and adds stability, the missing-orbit inventory, the extremal output maps and a minimal order-15 recurrence. Part V, added in batch 67, proves the formula for every  $k \geq 4$  once  $n \geq T(k) = 4k \log k + O(k)$  instead of  $n \geq 3k^4$  (so for all  $k$  up to  $cn/\log n$ ,  $c < 1/4$ ), with a theta-function limit of the optimal deficit when  $n/k^2$  tends to a constant and a Gaussian law in between. Part VI, added in batch 67, determines the minimal recurrences of the explicit witness deficit for every fixed  $k$ : order 9 at

three outputs (against Part II's 12), a product-12 cancellation on the odd residues but not in the full sequence at nineteen outputs, the only product-12 cancellation at any  $k$ , and explicit tail numerators.

## 22. Synchronizing Dzyga's automata by a cyclic retract

[automata-and-formal-languages/dzyga-synchronizing-retract/article.pdf](#)  
 from [Dzyga\\_Automata\\_Cyclic\\_Retract\\_Research.zip](#)

For the Figure-3-complete automaton  $A_k$  of Conjecture 8 in Szykula's 2026 survey, exhibits an explicit word  $R_k = p(ec^{r-1})^{r-2}e$  with  $r = k + 4$  that sends every state to  $q_2$  for every  $k \geq 1$ , of expanded length  $8k^2 + 54k + 92 = \frac{8}{9}n^2 + \frac{22}{3}n + 16$  for  $n = 3k + 6$ . The proof is a cyclic retraction with two projected reflections whose discrepancy supplies one adjacent merger, and does not depend on the finite computations; shortening the reflected tail by one state gives a nonsynchronizing automaton with a dihedral permutation quotient. This settles the synchronization clause of the stated model, not the numerical threshold conjecture, and the universal lower bound  $\tau(q_0) \geq 4k + 8$  is not proved.

## 23. Gaps in bounded-shuffle hierarchies

[automata-and-formal-languages/insertion-degree-spectra/article.pdf](#)  
 from [shuffle\\_spectra\\_research.zip](#); batch 41: [ProveIt\\_Binary\\_Insertion\\_Spectra.zip](#); batch 66: [ProveIt\\_Sharp\\_Insertion\\_Isolation.zip](#)

A counterexample to the insertion-degree No-Gap Conjecture of Hughes, arXiv:2608.27755v1, Section 11, followed by positive results: every finite set of positive integers containing 1 is the insertion-degree spectrum of two nonempty finite languages over a suitable alphabet, and any prescribed finite list of degrees above 1 is realizable with every other output of degree 1. A ternary pair has spectrum  $\{1, 3\}$  with sole difficult output **abcba**, whose length 5 is the minimum possible for a gap witness; a binary pair has spectrum  $\{1, 2, 4\}$  while all 18 singleton source-pair spectra are initial intervals, so source-pair ambiguity can erase a degree. The separate iteration-depth no-gap conjecture is proved for arbitrary languages. Part II, added in batch 41, fixes the alphabet: every finite set containing 1 is the spectrum of finite *binary* languages with the single inserted word  $a^m$ , so two letters always suffice; its  $m^2$  length bound holds in its full-coverage class only. Part III, added in batch 66, determines that class exactly: an isolated degree- $m$  output with  $g$  gaps between its  $a$ 's (the two end gaps included) and  $T$  letters  $a$  exists exactly when  $g \geq m$  and  $T \geq \max(g(m-1) + 1, \tau_m)$ , with  $\tau_3 = 12$  and  $\tau_m = 5m - 2$  for  $m \geq 4$ , so the minimum lengths for  $m = 3, 4, 5$  are 14, 21 and 27. It characterizes the extremal centers and the mandatory targets, and at  $m = 3$  classifies the shortest target languages:  $3^9 = 19,683$  of them, of minimum size 37, attained by 512.

## 24. Constrained crossover closure

[automata-and-formal-languages/constrained-crossover-closure/article.pdf](#)  
 from [ProveIt\\_Constrained\\_Crossover.zip](#) (batch 44); batch 66: [ProveIt\\_Crossover\\_Stabilization.zip](#), [ProveIt\\_Crossover\\_Classification.zip](#)

Answers the question Hughes (arXiv:2608.27755v1, Section 4) leaves open on the complexity of deciding  $L \otimes_c L = L$  for the constrained crossover of a context-free language: it is co-r.e.-complete. Adds an aligned-rank no-gap theorem for iterated crossover and effectivity results in which Presburger quantifier elimination is used only as an algorithmic step. Related to, but using nothing from, the insertion-degree report above. Part II, added in batch 66, shows finite stabilization of regular seeds PSPACE-complete for NFA input, already over two letters; the upper bound also covers DFA input, improving Part I's EXPSpace bound, while the exact DFA complexity stays open; in the stabilizing case the rank is at most quadratic in the number of states. Part III, added in batch 66, decides regularity, inclusion and equality of the full closures of context-free seeds, classifies the finitely stabilizing sparse one-marker seeds into regular, linear non-regular and non-context-free closures, and shows that thresholding the exact stabilization depth is coNP-complete for succinct ray input.

## 25. A cut-sensitive attack on the first open MIX grammar case

[automata-and-formal-languages/mix-grammar-cut-sensitive/mix\\_grammar\\_report.pdf](#)  
from [mix\\_grammar\\_research.zip](#)

The selected question — whether  $L_3 = \text{MIX}$  for the canonical tuple grammars of Yuyama, *On the Kanazawa–Salvati Conjecture* (arXiv:2609.13871v1) — is *not* settled here, and the report says so at the front. What it does contain, with complete written proofs, is a cut-sensitive membership recurrence, a broad three-component positive family, an infinite family in  $L_3 \setminus L_2$ , and an exhaustive computation through length 21.

## 26. Natural densities of the Tribonacci word’s additive complexities

[automata-and-formal-languages/tribonacci-additive-complexity/article.pdf](#)  
from [tribonacci\\_additive\\_density\\_research.zip](#)

Answers Remark 19 of Popoli, Shallit and Stipulanti (arXiv:2410.02409). With  $\beta$  the Tribonacci constant, the additive complexity of the Tribonacci word takes values in  $\{3, 4, 5\}$  with natural densities  $d_3 = (21\beta^2 - 38\beta + 5)/22$ ,  $d_4 = (-55\beta^2 + 72\beta + 67)/22$  and  $d_5 = (17\beta^2 - 17\beta - 25)/11$ , all positive and summing to 1, with a power-saving error  $O(N^\theta)$ ,  $\theta = \log(7/5)/\log \beta$ , at *every* integer cutoff rather than only along Tribonacci cutoffs. The 76-state output automaton is reconstructed independently rather than copied, and certified by evaluating its degree-76 characteristic polynomial at the adjacency matrix.

# 5 Hankel determinants and continued fractions

## 5.1 Somos sequences and elliptic curves

## 27. A proof of Barry’s Hankel–Somos-6 conjecture

[hankel-determinants/somos-and-elliptic/barry-somos-6-hankel-recurrence/article.pdf](#)  
from [barry\\_somos6\\_research.zip](#)

Conjecture 7 of Paul Barry, arXiv:2211.12637: for the Hankel determinants  $H_N$  of a four-parameter Jacobi-type generating function,  $H_N H_{N-6} = \alpha H_{N-1} H_{N-5} + \gamma H_{N-3}^2$  with  $\alpha = t^2(r+2)^2$  and  $\gamma = t^3((s+t)(r+s+t+2)^2 + t(r+2)^3)$ . Also proves a homogeneous four-parameter form and a seven-parameter polynomial Hankel theorem making exceptional specialisations rigorous; the verifier never divides by a determinant.

## 28. Finite Hankel Certificates for Three Somos-8 Conjectures

[hankel-determinants/somos-and-elliptic/somos-8-hankel-certificates/article.pdf](#)  
from [somos\\_hankel\\_proofs.zip](#)

Conjectures 11, 12 and 13 of the same Barry paper, by importing Hone’s genus-two Hankel theorem (arXiv:1907.05204v3, Thm. 5.4) and supplying a binomial translation, a polynomial-minor specialisation valid on the degenerate locus, and a rank-bound lemma that turns four symbolic rows into an all-index identity. Identifies a sign error in the source’s Catalan expression for Conjecture 13.

## 29. Elliptic addition and Jacobi continued fractions

[hankel-determinants/somos-and-elliptic/elliptic-addition-and-jacobi-fractions/article.pdf](#)  
from [elliptic\\_continued\\_fraction\\_research.zip](#)

A consistently indexed, non-torsion version of Barry’s 2023 Conjecture 4, with the coordinate formulas of Conjecture 3: every continued-fraction tail is given explicitly and removing one layer is shown to be elliptic translation. Hypotheses: characteristic zero,  $Y^2 + aXY + bY = X^3 + cX^2 + dX$  nonsingular,  $b \neq 0$ , and  $P = (0, 0)$  of infinite order; order-three and order-four examples show the regular fraction genuinely needs the last one.

## 5.2 Catalan and ballot moments

### 30. Shifted Catalan Hankel Polynomials

[hankel-determinants/catalan-and-ballot/shifted-catalan-hankel-polynomials/article.pdf](#)  
from [catalan\\_hankel\\_conjecture\\_solution.zip](#); batch 39: [ProveIt\\_Catalan\\_Root\\_Collisions.zip](#);  
batch 42: [ProveIt\\_Cyclotomic\\_Catalan\\_Recurrences.zip](#); batch 53:  
[Two\\_Endpoint\\_Catalan\\_Confluence.zip](#); batch 55: [ProveIt\\_Interior\\_Collision\\_Research.zip](#); batch 57:  
[ProveIt\\_Catalan\\_Growing\\_Shifts.zip](#)

For  $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})$ , proves the denominator-exponent assertion on page 4 of Barry, arXiv:2011.10827v1, and shows the separately printed Conjecture 2 needs an explicit one-unit shift correction. The minimal characteristic polynomial is  $(X^2 - (a+2b)X + b^2)^{m(m-1)/2+1}$ , so the minimal order is  $m(m-1) + 2$ ; all exceptional reduced denominators, the exact coefficient formula, numerator degree and reciprocal symmetry are supplied. Part II, added in batch 39, replaces  $aC + bC'$  by an arbitrary polynomial multiplier and computes the root collisions: exact sector degrees with nonzero leading constants, the minimal recurrence order off resonance, a secondary-Hankel factorization and a certified root-free algorithm for the recurrence order. Part III, added in batch 42, gives every characteristic multiplicity for one repeated root at a cyclotomic resonance, and shows that the question it answers needed one more parameter: the sign of  $z^L$ , not only  $t$  and  $\text{ord}(z^2)$ . Part IV, added in batch 53, answers the endpoint half of Part II's question on uniform limits: for fixed-size clusters of roots approaching the endpoints 0 and 4 ( $d$  roots in all) the normalized determinant is a convergent series in even powers of  $1/(N + d/2)$ , locally uniform through every collision inside a cluster, with a factorized leading profile, an explicit first interaction term, and ratio to the fixed-endpoint determinant tending to 1 exactly when the outward displacements are  $o(N^{-2})$ . Part V, added in batch 55, answers the other half: for fixed-size clusters of roots colliding on the scale  $1/N$  at fixed bases other than 0 and 4, the determinant is an exact finite sum of exponential sectors whose amplitudes are convergent series in  $1/(N + d/2)$ , uniform through every collision, with an explicit first correction (this expansion is not even), an explicit remainder bound, and factorized amplitudes for several separated clusters. For two equal-multiplicity roots inside the spectrum the limit is an oscillatory Jacobi Hankel determinant, the first correction oscillates as  $\sin(2(N+t)\theta)$ , and simple zeros move by a convergent phase-dependent expansion. Limits uniform up to the endpoints remain open. Part VI, added in batch 57, lets the shift grow with the determinant size: for real  $a \geq 0$  the ratio  $D_N^{(m)}(a, 1)/D_N^{(m)}(0, 1)$  tends to 1 exactly when  $aN(N+m+1)/(4m+2) \rightarrow 0$ , at any relative growth of  $N$  and  $m$ ; the size-biased inverse Jacobi zeros follow a shifted Marchenko–Pastur (Catalan-moment) law, and three critical scales, an algebraic free energy and a uniform saddle prefactor describe the crossover.

### 31. Product formulas for ballot-polynomial Hankel determinants

[hankel-determinants/catalan-and-ballot/ballot-polynomial-hankel-determinants/article.pdf](#)  
from [Cigler\\_Hankel\\_Conjectures\\_13\\_15.zip](#)

Conjectures 13, 14 and 15 in Section 5 of Johann Cigler, arXiv:2111.14492v3. An explicit coefficient product gives the sign and exact support of the normalised shift- $k$  Hankel determinant of the ballot moments and positivity of every residual coefficient for  $n \geq 1$ ; the same products give the conjectured generating-function denominators, exact numerator degrees and reciprocal numerator identities. Adds strict log-concavity of the residual coefficients and their limiting binomial profile.

### 32. Rectangular Schur polynomials and Cigler's Conjecture 8

[hankel-determinants/catalan-and-ballot/cigler-conjecture-8-schur/article.pdf](#)  
from [cigler\\_conjecture8\\_research.zip](#)

A complete proof of Conjecture 8 in Cigler, arXiv:2111.14492v3, Section 4 (printed pp. 16–17), for the Hankel determinants of  $b_m(t) = \sum_r \binom{\lfloor m/2 \rfloor}{r} \binom{\lceil m/2 \rceil}{r} t^r$ , including the numerator-polynomial degree and reciprocity assertions. A different conjecture of that paper from the ballot-polynomial entry (Conjectures 13–15) and from the parity entry (Conjecture 16).

### 33. A parity factorization for Cigler's Conjecture 16

[hankel-determinants/catalan-and-ballot/cigler-conjecture-16-parity/article.pdf](#)  
 from `cigler_conjecture16_research.zip`; batch 56: `ProveIt_Cigler_Schur_Recurrences.zip`; batch 66:  
`ProveIt_Cyclotomic_Pole_Collapse.zip`

A complete proof of Conjecture 16 in Cigler, arXiv:2111.14492v3, Section 6 (pp. 21–22), on the Hankel polynomials, together with a sharp stabilization statement: the report also determines the first coefficient at which the conjectured stabilization fails. Part II, added in batch 56, proves the report’s experimental rectangular-Schur formula: with  $a = \lfloor k/2 \rfloor$  and  $b = \lfloor (k-1)/2 \rfloor$  the normalized determinant is  $s_{(n^a)}(1^a, t, (t^2)^b)$  for all  $k \geq 1$  and  $n \geq 0$ . It determines the exact minimal recurrence at each shift, with reduced denominator  $\prod_{q=0}^{k-1} (1 - t^q z)^{1 + \lfloor q(k-1-q)/2 \rfloor}$  over  $\mathbb{Q}(t)$  and at every real  $t \neq 0, \pm 1$ , proves Cigler’s denominators (84) and (86) minimal, and shows that the positive reciprocity sign printed in his (85), part of Conjecture 18, is false when  $k = 2a$  with  $a$  even; Conjecture 17 stays open. Part III, added in batch 66, determines the reduced denominator and the minimal recurrence at every complex parameter: at a primitive root of unity of order  $\ell \geq 3$  at most one residue class loses pole order, decided by an explicit parity test, and it loses one order except for a loss of three at primitive fourth roots when  $k \equiv 1 \pmod{4}$ . Its own README explains why this is a different problem from the other Cigler entries in this collection.

### 5.3 Arithmetic Hankel determinants

The numerator formula recorded as a conjecture in OEIS A122251, with A122249 as its binary special case. Two reports attacked it and proved it the same way, sharing even their worked counterexamples; they are merged here into one. The merged report sits at the category’s top level.

#### 34. A Proof of an OEIS Hankel-Numerator Conjecture

[hankel-determinants/a122251-numerators-and-denominators/report.pdf](#)  
 from `hankel_numerator_research.zip` + `hankel_oeis_conjecture_proof.zip`

For the Hankel determinant

$$H_n(m, a) = \det(1/(a + m(i + j)))_{0 \leq i, j \leq n},$$

the reduced numerator is  $\prod_{p|m} p^{n(n+1)v_p(m)+2\sum_{k \leq n} v_p(k!)}$ : independent of the coprime offset  $a$ , a perfect square, and supported on the primes dividing  $m$ . Taking  $m = p$  and  $a = 1$  settles the OEIS A122251 conjecture, with A122249 as the binary case. Also gives a closed form for the reduced denominator and all its local valuations, a sharp gcd theorem over the allowed offsets realised at just two explicitly constructed shifts, the extension to unequal row and column steps  $\det(1/(a + m_1 i + m_2 j))$ , recurrences, digit-sum asymptotics, and a Lambert-series generating function with a natural boundary. *Merged; the determinant is evaluated three independent ways, and a second proof of the numerator formula runs through an explicit inverse matrix and MacMahon’s boxed plane partitions.*

### 5.4 Growth laws and run statistics

#### 35. A proof of the Apéry Hankel growth conjecture

[hankel-determinants/growth-and-runs/apery-hankel-determinant-growth/apery\\_hankel\\_growth.pdf](#)  
 from `apery_hankel_research.zip`

For  $A_k = \sum_j \binom{k}{j}^2 \binom{k+j}{j}^2$  and  $D_n = \det(A_{i+j})$ , proves  $\log D_n = n(n+1) \log \frac{17+12\sqrt{2}}{4} + O(n)$  — the limit conjectured in the September 2026 revision of OEIS A228143 — using Edgar’s moment-density theorem. Also an explicit law for  $\det(A_{m(i+j)+r_n})$  with  $r_n/n \rightarrow \rho$ , and non-D-finiteness of the formal generating series of  $D_n$ .

#### 36. Binary Runs in Rueppel Hankel Determinants

[hankel-determinants/growth-and-runs/rueppel-binary-run-determinants/rueppel\\_hankel.pdf](#)

from rueppel\_hankel\_research.zip

For the Rueppel series  $r(x) = \sum_{j \geq 0} x^{2^j - 1}$ , the polynomial identity  $H_n(tx + 1/r(x^2)) = (-1)^{1+\varepsilon+\lfloor R(n)/2 \rfloor} (\varepsilon + (R(n) - \varepsilon)t^2)$ , where  $R(n)$  counts runs of equal bits in the binary expansion of  $n$  and  $\varepsilon$  is the low bit of the Gray code  $n \oplus \lfloor n/2 \rfloor$ . Settles statements 8, 10, 13 and 23 of Barry's 2021 paper (not the identically numbered ones elsewhere), with a correction.

## 6 Congruences and $p$ -adic valuations

### 6.1 Supercongruences

#### 37. Congruences for OEIS A348410 at every prime

congruences-and-valuations/supercongruences/a348410-all-prime-congruences/article.pdf

from a348410\_supercongruence.zip + A348410\_Supercongruence\_Research.zip

For  $A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}$  proves the odd-prime cubic supercongruence  $v_p(A(Np) - A(N)) \geq 3v_p(N) - \varepsilon_p$  with  $\varepsilon_3 = 1$ ,  $\varepsilon_p = 0$  for  $p \geq 5$ , and no coprimality condition;  $(\alpha, \beta) = (2, 1)$  is Bala's A348410 conjecture. Separately proves an exact binary leading-defect formula,  $A(m2^r) - A(m2^{r-1}) \equiv 2^{3r-3} \alpha \beta \binom{(\alpha+\beta+1)m-1}{m-1} \pmod{2^{3r-2}}$ , hence the all-prime bound  $3r - v_p(12)$  and sharp denominators 12 and 24. An appendix audits arXiv:2104.10754v1, whose framing conclusion and weighted harmonic assertion fail on its own displayed examples. *Merged; the odd-prime and binary halves run on different mechanisms — a doubling automorphism, and reflection with formal integration by parts — and both are kept.*

**Not a duplicate of *signed-binomial-supercongruences*, and checked.** That report's  $a_{r,s,t}(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$  specializes at  $t = 1$ ,  $(r, s) = (-\beta, -\alpha)$  to the coefficient studied here, and the two odd-prime statements then agree. Neither contains the other. That one is general in the extraction step  $t$ , which is fixed at 1 here; this one adds the prime 2 with an exact defect and its sharpness, every prime at once, minimal common denominators, the primitive-cyclic-word interpretation, the Lambert/Euler product, the algebraic generating functions, a spacing-three counterexample, and a sufficient criterion beyond the family. The proofs are different arguments: that one imports Jacobsthal's congruence, this one uses none and pairs a quadratic-tail estimate with a coefficient estimate.

#### 38. Signed binomial supercongruences and OEIS A352373

congruences-and-valuations/supercongruences/

a352373-signed-binomial-supercongruences/signed\_binomial\_supercongruence.pdf

from OEIS\_A352373\_supercongruence.zip

For  $a(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$  proves  $a(mp^h) \equiv a(mp^{h-1}) \pmod{p^{3h}}$  for  $p \geq 5$  and  $a(m3^h) \equiv a(m3^{h-1}) \pmod{3^{3h-1}}$ , with  $m$  not required coprime to  $p$ . Covers A352373, A348410, A351856 and A351857 as specialisations; the only non-elementary import is the classical Jacobsthal binomial congruence.

**Not a duplicate of *all-prime-congruences*, and checked.** Setting  $t = 1$  and  $(r, s) = (-\beta, -\alpha)$  turns the theorem below into that report's odd-prime tower, and the moduli agree after reindexing the multiplier. Neither contains the other: this one is general in the extraction step  $t$ , that one is fixed at  $t = 1$  but proves the prime 2 case, the all-prime bound, the sharp denominators and much else. The two proofs are genuinely different — this one imports Jacobsthal's congruence, that one does not use it at all.

#### 39. Apéry numbers under exponentiation and series reversion

congruences-and-valuations/supercongruences/

a267220-apery-transform-supercongruences/article.pdf

from Apery\_OEIS\_Supercongruences.zip, apery\_oeis\_conjectures(1).zip

Proves both supercongruence conjectures contributed by Peter Bala to OEIS A267220 on 17 October 2024, for every integer parameter including negative and zero. With  $A(x) = \exp \sum_{n \geq 1} a_n x^n / n$

for the Apéry numbers  $a_n$ ,  $F(x) = x/\text{Rev}(xA(x))$ ,  $u_m(n) = [x^n]A(x)^{mn}$  and  $v_m(n) = [x^n]F(x)^{mn}$ : for odd  $p$  and  $R = \nu_p(N)$  both families satisfy  $\nu_p(u_m(N) - u_m(N/p)) \geq 2R + \nu_p(m)$ , the unsigned statement for  $v_m$  holds at *every* prime including 2, and a *signed* refinement  $(-1)^{mN}u_m(N) \equiv (-1)^{mN/p}u_m(N/p)$  recovers full precision at  $p = 2$ , where the unsigned bound genuinely loses one power. Merged from two reports that share a seed lemma and a Möbius/dilogarithm layer but diverge at the crux; both routes are kept, and the merged text is careful that the signed and unsigned 2-adic statements are different statements, not a sharper and a weaker version of one.

#### 40. Four Motzkin congruences

[congruences-and-valuations/supercongruences/a001006-motzkin-congruences/article.pdf](#)  
from [motzkin\\_congruences.zip](#)

Proves all four conjectures in Peter Bala's comment of 10 February 2022 on OEIS A001006, with sharp ranges. The engine is a prime-power block-transfer theorem: for odd  $p$  and  $q = p^{k-s+1}$ ,  $h = mp^{s-1}$ , the Motzkin numbers satisfy  $M_{mp^k+d} \equiv T_h M_d$  for  $0 \leq d \leq q-3$ , with explicit corrections at  $d = q-2$  and  $d = q-1$ ; the boundary case gives  $M_{np^{r-2}} \equiv \frac{\varepsilon_p^{r-1}}{2} T_{n-1} - U_{n-1} \pmod{p}$ . Adds the exact first failures beyond the two initial-block ranges, arbitrary-multiplier and all-prime-power boundary extensions, a prime-modulus digit algorithm, and examples showing why unweighted repetition modulo a prime cube is generally false. Discusses Burns, Kohen and Pan-Sun rather than claiming novelty.

### 6.2 Congruences for iterated and compositional series

#### 41. Arithmetic of a Self-Iterating Generating Function

[congruences-and-valuations/iterated-series/a396807-self-iterating-series/article.pdf](#)  
from [A396807\\_congruences\\_and\\_prime\\_power\\_lifts.zip](#)

Both conjectures in OEIS A396807 for  $A(x) = x + A^{[5]}(x)A^{[6]}(x)$  (functional iteration):  $A(x) \equiv x/(1-x)$  and  $A^{[k]}(x) \equiv x/(1-kx)$  modulo 10 for every integer  $k$ , negative iterates included. Adds a sharp modulus theorem for  $F = x + cF^{[r]}F^{[s]}$ , rational lifts at every selected prime power, and minimal period 46860 for A396807 modulo 100 starting at index 1.

#### 42. Prime-power congruences for compositional tree series

[congruences-and-valuations/iterated-series/compositional-tree-series-congruences/article.pdf](#)  
from [compositional\\_tree\\_congruences.zip](#)

For  $A_l(x) = x \exp(A_l^{[s]}(x))$ , shows the OEIS A396805 claim  $a_5(n) \equiv n \pmod{5}$  is *false*: first failure  $n = 23$ , with exception set  $\{20j+3, 20j+4 : j \geq 1\}$  and residue 2 throughout. Proves eleven other conjectural assertions in A396803–A396806 via a general reduction of coefficients mod  $p^\alpha$  to typed rooted trees on fewer than  $p^\alpha$  labels, yielding eventual periods, rational residue generating functions, and an exact classification of  $a_l(n) \equiv n \pmod{p}$ .

### 6.3 Valuations, divisibility and periodicity

#### 43. Binary Carries and the Divisibility of Two-Stack-Sortable Permutation Numbers

[congruences-and-valuations/a000139-binary-carries-parity/research.pdf](#)  
from [a000139\\_carries\\_research.zip](#)

Proves Peter Bala's parity conjecture for OEIS A000139,  $a(n) = 2(3n)!/((n+1)!(2n+1)!)$ :  $a(n)$  is odd exactly when  $n$  is odd and its binary expansion contains no 11. The full valuation is  $v_2(a(n)) = s(n) + s(n+1) - s(3n)$  in binary digit sums. Two independent parity proofs, plus exact valuation distributions, recurrences, asymptotics, extremal formulas and limit laws.

#### 44. Polynomial divisibility of height-one factorial ratios

[congruences-and-valuations/factorial-ratio-polynomial-divisibility/article.pdf](#)

from bala\_factorial\_divisibility\_research.zip

A general rational-root criterion for uniform polynomial divisibility of balanced integral factorial ratios, applied to Peter Bala’s uniform-product conjectures for OEIS A211417 and A295431: explicit multipliers for every product length, with the least multipliers certified in eight particular examples. Includes an atlas of the 52 sporadic height-one ratios with recurrences, generating functions and asymptotic expansions.

#### 45. When the Secant Numbers Become Periodic

[congruences-and-valuations/secant-number-periodicity/article.pdf](#)

from secant\_periodicity\_research.zip

Refutes the OEIS A000364 conjecture that the secant numbers from  $n = 1$  are purely periodic modulo every  $m$ : at  $m = 27$ ,  $a(1) \equiv 1$  but  $a(19) \equiv 10$ . The exact replacement is that pure periodicity from index one holds if and only if no odd prime cube divides  $m$ ; the least eventual period always divides  $\varphi(m)$ . Determines the least eventual period and exact onset for every modulus, with densities and a certified mean onset.

## 7 Tetration and digit stabilization

#### 46. A Counterexample to a Decimal-Length Bound for Tetration Stabilization

[tetration-and-digit-stabilization/congruence-speed-decimal-length-bound/article.pdf](#)

from tetration\_counterexample.zip

An explicit 2544-digit base  $A = 1 + qu$  ( $q = 5^{2547}$ ) for which  $\nu_{10}(T_{n+1} - T_n) = \min(1 + 2548n, 2547(n + 1))$  at every height, so the congruence speed at height  $\text{len}(A) + 2$  is 2548 while the permanent value is 2547. This contradicts the decimal-length form of Conjecture 1 of Marco Ripà, *The congruence speed formula* (NNTDM 27(4), 2021). The report audits the later Ripà–Onnis confirmation: their weaker valuation bound is compatible with the counterexample, the decimal-length bound is not.

#### 47. Tetration, perfect powers, and frozen decimal digits

[tetration-and-digit-stabilization/perfect-power-minima-for-frozen-digits/article.pdf](#)

from tetration\_perfect\_power\_minima.zip

Determines  $\rho(n)$ , the smallest  $n$ th perfect power  $x^n > 1$  whose eventual number of newly frozen terminal decimal digits is exactly  $n$ :  $\rho(1) = 2$ ,  $\rho(2) = 49$ , and  $\rho(n) = B(s)^n$  for  $n \geq 3$  with  $s = n - v_2(n)$  and  $B(s) \in \{2^s \pm 1, 3 \cdot 2^s \pm 1\}$  according to  $s \bmod 4$ . All other roots are ruled out by a deterministic quadratic size barrier plus small exact certificates.

#### 48. Phase-shift classification for integer tetration

[tetration-and-digit-stabilization/](#)

[phase-shift-classification/tetration\\_phase\\_classification.pdf](#)

from tetration\_phase\_classification.zip

Proves the 21-word phase-shift conjecture in OEIS A376842, the 14-word even-base restriction from the appendix of Ripà’s 2025 paper, and the 28-word prescribed-anchor conjecture in OEIS A376446. The mechanism compares the 2-adic and 5-adic valuations of  $D_b = T_{b+1} - T_b$  at the first permanently constant-speed height: strict 2-adic dominance gives phase 5, strict 5-adic dominance a multiplicative orbit in  $\mathbb{F}_5^\times$ , and balanced valuations exactly four further anchored words.

#### 49. A Sharp Convergence Theorem for the $q$ -Newton Series of Subcritical Tetration

[tetration-and-digit-stabilization/q-newton-series-convergence/article/](#)

[tetration\\_qnewton.pdf](#)

from tetration\_qnewton\_research.zip

Proves convergence of the explicit  $q$ -binomial interpolation series of MathOverflow question 259278 for every real base  $1 < a < e^{1/e}$ , with  $\log a = qe^{-q}$ : the grouped series converges absolutely and locally uniformly for  $\text{Re } z > -2$ , converges conditionally on  $\text{Re } z = -2$  except at

$z = -2 + 2\pi ik/(-\log q)$ , and diverges for  $\operatorname{Re} z < -2$ . Adds strict complete monotonicity and uniqueness under an eventual log-convexity condition.

#### 50. Stabilization of integer power towers

tetration-and-digit-stabilization/power-tower-stabilization/article.pdf  
from tetration\_stabilization\_research.zip + tetration\_exact\_digits.zip

Refutes the claim that every prime base ending in 9 reaches its permanent decimal congruence speed by height 2 (Ripà, Conjecture 2; arXiv:2208.02622v1, Conjecture 4.1): 2749 is the smallest prime counterexample, with  $H(2749) = 3$ . Proves exact formulas for  $C_a(h)$ ,  $V(a)$  and  $H(a)$ , an exact natural density for onsets, and infinitely many primes at each prescribed onset via CRT and Dirichlet. Also resolves the three conjectural comments in OEIS A324017 — two proved, the third disproved, with  $3 \uparrow\uparrow 2 \equiv 27$  and  $3 \uparrow\uparrow 3 \equiv 59 \pmod{64}$  — via  $T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}$  for even  $q \geq 4$ , so exactly one radix- $q$  digit freezes per height. Adds a factorization-free digit-lifting algorithm, modular saturation across Knuth arrow ranks, and a sign-sensitive local  $p$ -adic Lambert  $W$  formula. *Merged; the shared one-digit-per-height theorem was proved identically in both originals and is proved once here.*

#### 51. Delayed digit stabilization: a counterexample and sharp bounds

tetration-and-digit-stabilization/  
iterated-power-digit-stabilization/delayed\_digit\_stabilization.pdf  
from digit\_stabilization\_research.zip

For  $S_a(b) = v_{10}(a^{10^{b+1}} - a^{10^b})$  and  $D_a(b) = S_a(b) - S_a(b-1)$ , answers negatively the question whether  $D_a(b) = 1$  for all  $b \geq 3$ : the smallest counterexample is  $a = 3 \cdot 2^{99} + 1$ , with  $S_a(2) = 100$ ,  $S_a(3) = 102$ ,  $D_a(3) = 2$ . Proves minimality, an exact onset formula, the smallest counterexample at every stage, the natural density of exceptions, and arbitrary-radix extensions. The article is explicit that  $a^{10^b}$  is not genuine tetration and derives the tower formulas separately.

## 8 Log-concavity, unimodality and positivity

#### 52. Single curves do not force log-concavity

log-concavity-and-unimodality/cluster-variable-log-concavity-counterexample/article.pdf  
from annular\_single\_lamination\_counterexample.zip

An unpunctured annulus with one marked point on each boundary and a single unit-weight simple lamination curve gives, after mutations 2, 1, 2 with the two mutable variables specialised to 1, the polynomial  $1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6$ ; the consecutive coefficients 3, 2, 4 form a strict valley, refuting Conjecture 1 of arXiv:2508.04396v3 and conflicting with its general unpunctured-surface unimodality statement. The geometric realisation is proved, not merely the matrix calculation, and a fixed seed gives infinitely many such examples.

#### 53. Cycles refute infinite log-concavity for chromatic coefficients

log-concavity-and-unimodality/infinite-log-concavity-of-cycles/article.pdf  
from chromatic\_cycle\_logconcavity.zip

For the cycle  $C_n$  with absolute chromatic coefficients under the iterated operator  $L(a)_k = a_k^2 - a_{k-1}a_{k+1}$ : infinitely log-concave for  $3 \leq n \leq 11$ ; first failure at depth 5 for  $n = 12$ , depth 4 for  $13 \leq n \leq 16$ , and depth 3 for  $n \geq 17$ . So  $C_{12}$  is the smallest cycle counterexample (no minimum over all graphs is claimed), with an explicit third-iterate witness polynomial valid for all  $n \geq 3$ .

#### 54. Oscillating Log-Concavity in Geometric Generating Functions

log-concavity-and-unimodality/oscillating-log-concavity-geometric-transform/article.pdf  
from oscillating\_log\_concavity\_research.zip

Challenges 2 and 3 in Section 3 of Heim and Neuhauser, arXiv:2302.13327v1, for the *geometric* transform  $1/(1 - uG(z))$ . For divisor-sum weights, every real exponent and every  $u > 0$  give infinitely many strict log-concavity and log-convexity indices with eventually bounded gaps; likewise for integer power weights with exponent  $\geq 2$ , the exponents 0 and 1 being explicitly

exceptional. For square weights at  $u = 1$ , every four consecutive indices carry both signs (sharp), each with density  $1/2$  in every arithmetic progression. A subdominant-pole theorem is the common mechanism.

### 55. A counterexample and a sharp spacing-three criterion for products of $q$ -integers

[log-concavity-and-unimodality/q-integer-product-unimodality/report.pdf](#)

from [q\\_unimodality\\_attack.zip](#)

The necessity half of Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko and Yang (arXiv:2605.12822v1) is false:  $(1+q)^6(1+q^3)$  is symmetric and weakly unimodal although no  $a_i$  is divisible by 3 and  $b = 2$  exceeds the asserted bound, and the example lies inside the source's claimed range. The report also proves the proposed sufficient condition for every  $r \geq 2$ , a necessary degree bound in the absence of an  $r$ -divisible ordinary factor, and a complete classification of spacings two and three.

### 56. Sharp size thresholds for failures of log-concavity in independence systems

[log-concavity-and-unimodality/independence-system-thresholds/article.pdf](#)

from [sharp\\_log\\_concavity\\_thresholds.zip](#)

For each rank  $r \geq 4$ , determines the smallest ambient ground-set size admitting failure of ordinary, ordered and ultra log-concavity, and shows the three thresholds coincide across all independence systems, those satisfying Xie–Xu's augmentation bound, and those with additive augmentation parameter 3 (values  $7/6/5$ ,  $7/7/6$ ,  $8/7/7$ , and  $r+1$  throughout for  $r \geq 7$ ). The initial target, Xie–Xu Conjecture 4.8 (arXiv:2408.09152v1), had already been disproved by Alex Chengyu Li in September 2026; the report states this and refines the target to the rankwise minimum-size question.

### 57. A sharp order-five threshold for higher-order Stirling subset log-concavity

[log-concavity-and-unimodality/higher-order-stirling-rows/article.pdf](#)

from [stirling\\_log\\_concavity.zip](#)

For  $S_{n,k}^{(r)}$ , the number of partitions of  $n + (r-1)k$  labels into  $k$  blocks of size  $\geq r$ , proves that all rows are log-concave if and only if  $1 \leq r \leq 5$ , with strict inequalities in the interior — the log-concavity assertion of Deb–Sokal Conjecture 1.4(c) (arXiv:2507.18959v1). Also proves the cycle cases  $r = 3, 4$ ; the fifth-order *cycle* conjecture is explicitly left unresolved, and the included operator counterexample is flagged as not being a counterexample to it.

### 58. A sum-of-squares proof of Ulas's real-zero conjecture

[log-concavity-and-unimodality/stern-polynomial-positivity/stern\\_conjecture\\_note.pdf](#)

from [stern\\_conjecture\\_research.zip](#)

Conjecture 4.4 of Maciej Ulas, arXiv:1909.10844v1, for the Stern polynomials indexed by  $a_n = (4^n - 1)/3$  and  $h_n = 2(4^n - 1)(2 \cdot 4^n + 1)/3 + 1$ : for every  $n \geq 0$  and every real  $t$ ,  $W_n(t) = (A_{n+1}(t) - tA_n(t)/2)^2 + 3t^2 A_n(t)^2/4 > 0$ . The adjacent-polynomial quadratic identity behind it was already in Ulas's paper and is re-derived, not claimed as new.

### 59. Two-coin counterexamples to generalized Shepp–Olkin thresholds

[log-concavity-and-unimodality/bernoulli-entropy-quasiconcavity/article.pdf](#)

from [bernoulli\\_entropy\\_research.zip](#); batch 43: [proveit\\_product\\_entropy.zip](#); batch 57: [ProveIt\\_Alphabet\\_Entropy.zip](#)

For every finite real  $q > 1$ , both the Rényi- $q$  and the Tsallis- $q$  entropy of a sum of independent Bernoulli variables fail even *quasiconcavity* as a function of the parameter vector, already on the open square  $(0, 1)^2$  with two coins. This disproves the proposed thresholds — Rényi 2 and Tsallis 3.65986... — adjoined to Conjecture 4.2 of arXiv:1503.01570 and restated in arXiv:1510.05390, Section 8. Part II, added in batch 43, settles the subunit joint-concavity question for the joint bit vector only: product Rényi entropy is concave in every dimension for  $0 < q \leq 2$ , and product Tsallis entropy is concave for every  $0 < q < 1$  exactly when  $n \leq 9$ . Part III, added in batch 57, answers Part II's question on larger alphabets for full categorical simplices: for  $0 < q < 1$  product Tsallis entropy on  $\Delta_{k_1} \times \cdots \times \Delta_{k_n}$  is concave exactly when  $\sum_i \Theta_{k_i}(q) \leq 1$  for an explicit alphabet

profile  $\Theta_k$ ; at  $q = 1/2$  three factors with a common alphabet of  $k$  symbols are concave exactly when  $k \leq 17$ ; concavity for every choice of  $n$  alphabets holds exactly when  $nq \leq 1$ ; and for  $1 < q < 2$  concavity reduces to positive semidefiniteness of an  $n \times n$  compression matrix. For the ordinary sum Part I's question stays open; an external claim about it is recorded as unverified.

## 60. Nonreal roots survive in polynomial iterative equations

[log-concavity-and-unimodality/nonreal-roots-in-iterative-equations/article.pdf](#)

from `nonreal_roots_research.zip`; batch 41: `ProveIt_Increasing_Spectral_Core.zip`; batch 42:

`ProveIt_Orientation_Reversing_Spectra.zip`; batch 54: `ProveIt_Exact_Algebraic_Degree.zip`

Here  $P[f] = \sum_j a_j f^{oj}$  means a combination of *iterates*, not a polynomial in pointwise values. Gives a negative answer to Problem 6.1 of Draga–Morawiec (arXiv:1503.00570), which proposed deleting nonreal maximal-modulus characteristic roots: an explicit odd, strictly increasing, globally bi-Lipschitz homeomorphism realises a cubic whose nonreal root cannot be discarded. Adds a classification of the circular root spectra that occur. Part II, added in batch 41, classifies the minimal iterative polynomials of increasing homeomorphisms with roots on several circles, through a universal spectral core and an affine rigidity theorem. Part III, added in batch 42, does the same for decreasing homeomorphisms with  $P(0) \neq 0$ : two-cycle and one-sided spectra, a decreasing core with one or two witnesses, and a finite-smoothness classification. Part IV, added in batch 54, answers Part III's question on algebraic degree: for a polynomial coordinate  $H$  and  $f = H \circ (a \text{ id}) \circ H^{-1}$  the degree of  $f$  over  $\mathbb{R}(x)$  is  $\deg H / \deg W$ , where  $W$  is the largest normalized right compositional factor of  $H$  with  $W(at) = a^{\deg W} W(t)$ , and it is exactly  $\deg H$  for every coordinate of Part III's finite-smoothness theorem. For every odd  $n \geq 5$  it gives real-analytic involutions of algebraic degree  $n$ , with distortion arbitrarily close to 1, whose graph extension has Galois group  $S_n$ , so they are not expressible by radicals.

## 61. Canonical binomial decomposition does not preserve real roots

[log-concavity-and-unimodality/binomial-decomposition-real-roots/article.pdf](#)

from `mu_welker_recursive_decomposition.zip` + `recursive_binomial_counterexample.zip`

A negative answer to Question 3.10 of Mu and Welker (arXiv:2503.24076v1) for their canonical decomposition  $F(t) = G(t) + tH(t)$  built from the integer binomial expansion of every coefficient: real-rootedness of  $F$  does not pass to the parts. Two cubic witnesses are carried, one with a repeated root and one with simple roots, each with its own digit table, discriminant pair and exact nonreal zeros; the report also gives the minimal witness within cubics, two all-degree thresholds neither of which implies the other, and a two-sided limit. *Merged; the two originals were two write-ups of one proof program, each already containing both witnesses with the headline and the co-star swapped.*

## 62. A first-triple criterion for log-concavity of MDS weight distributions

[log-concavity-and-unimodality/mds-weight-log-concavity/article.pdf](#)

from `mds_log_concavity_research.zip`

For  $d = n - k + 1 \geq 2$ ,  $k \geq 3$  and  $q > d$ , the compressed formal MDS weight distribution  $(1, A_d, \dots, A_n)$  is log-concave if and only if the single inequality  $A_{d+1}^2 \geq A_d A_{d+2}$  holds — the whole sequence is controlled by its first triple.

## 63. A uniform Hankel obstruction for higher-order Stirling subset polynomials

[log-concavity-and-unimodality/stirling-hankel-obstruction/article.pdf](#)

from `stirling_hankel_obstruction.zip`

Proves the higher-order negative assertion of Deb–Sokal Conjecture 1.4(d), with a stronger statement at every positive shift: for  $r \geq 3$  and  $a \geq 1$  the  $3 \times 3$  Hankel determinant  $\det(s_{r,a+i+j}(x))$  of the row-generating polynomials fails the required positivity. Clause (d) of that conjecture; clause (c) is the subject of the higher-order Stirling rows entry.

## 9 Quaternionic analysis

One report, first merged from ten manuscripts and since extended by two more. The subject is new to this collection, and the delivery was unusual: one expository survey and nine independent attempts at the same follow-up problem, all written on the same day. The nine agree, which is the substance of the entry.

### 64. Slice regularity and the global inverse Fueter problem

[quaternionic-analysis/slice-regularity-and-fueter-inversion/article.pdf](#)

from quaternionic\_analysis.zip and nine companion archives; batch 39:

ProveIt\_Fueter\_Polynomial\_Periods.zip; batch 64: ProveIt\_Fueter\_Global\_Inversion.zip

One article, first assembled from ten manuscripts. Part I is the delivered expository survey: slice regularity and Cauchy–Fueter regularity side by side, with Cauchy formulas, Taylor and Laurent expansions, the regular product and the geometry of zeros, the argument principle, Rouché, open mappings, Schwarz–Pick and an intrinsic Riemann mapping theorem. Parts II–XI answer the follow-up question the other nine manuscripts all attacked independently: for an axially monogenic  $g = P + IQ$ , when is there a single-valued slice-regular  $f$  with  $\Delta_4 f = g$ ? The answer is a two-period obstruction. The closed one-forms  $\omega_g = \frac{1}{2}(-P dx + Q dy)$  and  $\theta_g = \frac{1}{2}((xP + yQ)dx + (yP - xQ)dy)$  have vanishing periods exactly when a global primitive exists, and then  $f(x + Iy) = (x + Iy)C + V$  with  $dC = \omega_g$ ,  $dV = \theta_g$ , unique modulo  $q \mapsto qa + b$ ; the monodromy is affine, and the cokernel of the Fueter map has real dimension  $8m$ . The point the nine exist to make is that  $\theta_g$  is computed from the target, with no first primitive chosen. **Nine independent derivations reaching the same two forms, the same  $8m$  and the same logarithmic energy constant is the evidence the entry rests on.** Where they differed it was almost always notation — thirteen conflicts are recorded and resolved in MERGE\_NOTES.md — but two were mathematical and would have produced a false statement if the merge had been mechanical: the size law at a critical singularity has genuinely different sharp constants in two norms that four manuscripts both called  $M_g$ , and the conserved flux recovering the slope residue needs the weight  $3(\operatorname{Re} q - u) + \operatorname{Im} q$  rather than  $q - u$ , the coefficient 3 being exactly what makes it right-Fueter-regular. All ten source manuscripts and all ten verification programs are kept; the deduplication is in the article. Parts XII–XIII, added in batch 39 from an eleventh manuscript, answer the report’s open problem on higher dimensions in large part: in ambient dimension  $2h + 2$  one polynomial closed form of degree  $2h - 1$  (at  $h = 1$  the form  $\theta_g + t\omega_g$ ) gives a complete obstruction, with a constructive inverse, a cokernel of dimension  $2h d_V m$ , an exact residue filtration and sharp size and energy constants. Part XIV, added in batch 64 from a twelfth manuscript, carries this across the real axis and to planar domains of arbitrary, also infinite, connectivity: every admissible polynomial period character is realized, a continuous linear right inverse of the period map exists exactly when the admissible period rank is finite, and a continuous nonlinear section exists always.

## 10 Graph theory

### 65. Mutual-visibility sets in Andrásfai graphs

[graph-theory/andrasfai-mutual-visibility/article.pdf](#)

from andrasfai\_visibility\_research.zip

Proves the generating function attributed to Andrew Howroyd in OEIS A391632,  $\sum a(n)x^n = x(4 - 19x + 33x^2 - 40x^3 + 8x^4)/(1 - 10x + 29x^2 - 32x^3 + 8x^4)$ , by an exact weight-preserving bijection between mutual-visibility sets of the  $n$ -Andrásfai graph and anchored closed walks of length  $3n - 1$  in a four-state labelled graph. Adds the full bivariate generating function, an exact binomial formula for every cardinality, maximum-set classification, eventual polynomiality at each fixed deficit, exponential asymptotics, a central limit theorem, and linear-time recognition and exact uniform sampling.

### 66. A bicyclic 30-vertex graph with domination root $-4$

[graph-theory/domination-root-minus-four-order-30/domination\\_root\\_30.pdf](#)

from domination\_root\_30\_research.zip; batch 67: ProveIt\_Domination\_Cycle\_Obstructions.zip

An explicit connected planar graph on 30 vertices and 31 edges, treewidth 2 and domination number 10, with  $D(G, x) = x^{10}(x + 4)Q_{19}(x)$  — a negative answer to question 2 of Section 4 in Alikhani and Griswold, arXiv:2608.00109v1, which asks whether 33 is the minimum order. Their 33-vertex example had already disproved the original integer-root conjecture; no claim of global minimality at 30 is made. Three independent exact polynomial computations agree. Part II, added in batch 67, proves  $1 \leq |D(G, -1)| \leq 3^\beta$  sharply for cycle rank  $\beta$ , a simultaneous divisibility budget for integer roots and their multiplicities, and the exact value sets of  $|D(G, -1)|$ :  $\{1, 3, 7, 9\}$  for  $\beta \leq 2$ , so no root  $-6$  at cycle rank at most two, and a ten-element set for  $\beta \leq 3$  (computer-assisted). Part I's zero- $C$  four-cycle search forces  $|D(G, -1)| = 3$ , so it can produce no integer root below  $-2$  other than a simple  $-4$ ; no root  $-6$  or  $-8$  is constructed.

### 67. A marked-cycle proof of the trapezohedral domination conjecture

[graph-theory/trapezohedral-minimal-dominating-sets/article.pdf](#)

from trapezohedral\_domination\_research.zip, A381190\_proof\_package.zip

Proves the generating function labelled conjectural in OEIS A381190: for the  $n$ -trapezohedral graph, the number of dominating sets that are inclusion-minimal among all dominating sets and induce a connected subgraph is  $a_n = 2n(2p_{n-2} + p_{n-3})$  with  $p_m = p_{m-2} + p_{m-3}$  (Padovan). The proof classifies such sets by a cyclic arrangement of gaps of length 2 and 3 with a restricted marked edge. Adds the size distribution, strict log-concavity, an unbiased sampler, a Binet formula and a central limit theorem.

## 11 Enumerative combinatorics

### 68. A Twelve-Value Proof of the A181280 Formula

[enumerative-combinatorics/a181280-binary-matrix-formula/article.pdf](#)

from A181280\_research\_package.zip

The binary-matrix formula recorded as Conjecture 20 by Kauers and Koutschan (2023) and still marked conjectural in OEIS A181280: an explicit spectral recurrence bound and a finite certificate resting on twelve decisive counts, together with further consequences. STATUS.md records that general C-finiteness was already proved in Kauers's December 2025 lecture slides.

### 69. Esthetic Numbers, Folded Paths, and Algebraic Diagonals

[enumerative-combinatorics/esthetic-numbers-and-folded-paths/article.pdf](#)

from esthetic\_walks\_research.zip

The diagonal conjecture  $T(q, q - 2) = A182555(q - 2)$  in OEIS A377000, where  $T(q, k)$  counts base- $q$  positive integers of length  $k$  whose adjacent digits differ by one. Proves it exactly, proves the entry's other two conjectures as retrieved, and derives formulas for every fixed-offset diagonal  $q = k + d$ .

### 70. Diagonal Hardinian Arrays: A Fixed-Parameter Asymptotic Formula

[enumerative-combinatorics/hardinian-array-diagonal-asymptotics/article.pdf](#)

from hardinian\_arrays\_research.zip

For every fixed  $r \geq 0$ ,  $H_r(n, n) \sim C_r 4^{rn} n^{-r(r-1)/2}$  with an explicit  $C_r$  in Gamma values. The formula confirms the constants tabulated by Dougherty-Bliss and Kauers (2024) but refutes their proposed restriction to powers of 2, 3 and  $\pi$ :  $C_7 = 2^{18} 5^2 / (3^{45} \pi^3)$ . Adds an exact finite determinant at fixed array size, an exact enumeration algorithm, and a limiting negative-binomial law for total deficits.

### 71. An explicit formula for lucky parking spots

[enumerative-combinatorics/lucky-parking-spots/article.pdf](#)

from lucky\_parking\_spots\_research.zip

Proves Conjecture 13 of Butler, Hadaway, Lenius, Martens and Moats (JIS 29 (2026), Article 26.1.1): a closed form for  $C(n, j)$ , the number of parking functions in which *spot*  $j$  is lucky. Adds

an exact reflection law, all fixed right-edge columns, a closed form for OEIS A374533, Poisson edge limits, uniform bulk convergence, polynomial reciprocity and asymptotic expansions.

## 72. A hexagonal-lattice proof of the A195806 conjecture

[enumerative-combinatorics/a195806-hexagonal-lattice/article.pdf](#)  
from [A195806\\_hexagonal\\_proof.zip](#)

An elementary derivation of the six residue-class formulas and the constant-coefficient recurrence recorded as conjectural in OEIS A195806, starting from the original triangular-array constraints rather than from interpolation. The steps are an integral six-parameter bijection, four independent translation intervals, a dihedral chamber reduction, and an exact lattice count.

## 73. Leaves of large type in the numerical-semigroup tree

[enumerative-combinatorics/numerical-semigroup-leaf-types/article.pdf](#)  
from [leaf\\_type\\_counterexample.zip](#)

Disproves Conjecture 4.4 of Chappelon, Ramírez Alfonsín and Stamate, *Numerical semigroup tree: type-representation* (arXiv:2507.15006v2), with an explicit genus-34 counterexample, and establishes a sharp asymptotic order for the type of leaves in that tree.

## 74. Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

[enumerative-combinatorics/adjacency-bounded-132-avoiders/article.pdf](#)  
from [strict\\_growth\\_132.zip](#); batch 38: [Closing\\_Catalan\\_Growth\\_Gap.zip](#); batch 43: [ProveIt\\_Largest\\_Jump\\_Research.zip](#); batch 52: [ProveIt\\_Macroscopic\\_Jump.zip](#), [ProveIt\\_Macroscopic\\_Jump\\_Research.zip](#); batch 64: [ProveIt\\_Critical\\_Jump\\_Moments.zip](#); batch 68: [ProveIt\\_Polynomial\\_Rarity.zip](#)

For 132-avoiding permutations with every adjacent difference at most  $m$  and growth constant  $\alpha_m$ : proves  $\alpha_{m+1} > \alpha_m$  for all  $m \geq 1$  — the remaining strict-inequality part of Nadler’s Conjecture 2 — and the refinement  $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$ , with  $2 \log \pi - 4 \log 2 \leq \liminf E_m \leq \limsup E_m \leq 2 \log \pi$ . Part II, added in batch 38, closes that window: a Catalan-weighted supersolution and a growing cutoff give  $E_m \rightarrow 2 \log \pi$ , with an all-orders inverse-log expansion. Its numerics at  $m = 10^6$  (near 3.15) are far from the limit and are not evidence for it. The all- $m$  component ordering is still not claimed. Part III, added in batch 43, describes the joint limit near the boundary  $m = n - d$ : the distance  $n - M_n$  of the largest jump converges in total variation at rate  $n^{-1/2}$  to an explicit algebraic law, uniformly in  $d = o(n)$ . Part IV, merged in batch 52 from two manuscripts, treats the bulk regime  $n - m \asymp n$ :  $\sqrt{n} P(n - M_n > un) \rightarrow (3/\sqrt{\pi})(1 - 2u)/\sqrt{u(1 - u)}$  for  $0 < u < 1/2$  (and  $\rightarrow 0$  for  $u \geq 1/2$ ), so for  $1/2 < \theta < 1$  the fraction of 132-avoiders of length  $n$  with every adjacent difference at most  $\lfloor \theta n \rfloor$  is asymptotic to  $3(2\theta - 1)/\sqrt{\pi\theta(1 - \theta)n}$ , and the equivalent  $3/\sqrt{\pi(1 - \theta)n}$ , which Part III declined to assert, is false. It also gives the exact constant of every moment  $E(n - M_n)^p$ ,  $p > 1/2$  (for instance  $E(n - M_n) \sim 3\sqrt{n/\pi}$ ); for  $\theta \leq 1/2$  it proves only  $o(n^{-1/2})$  and polynomial lower bounds. Part V, added in batch 64, settles the critical moment,  $E\sqrt{n - M_n} = \frac{3}{2\sqrt{\pi}} \log n + c_\star + o(1)$  with an explicit  $c_\star \approx -0.9421$ , and gives the first correction to every subcritical moment. Part VI, added in batch 68, gives the order of Part IV’s fraction for every fixed  $0 < \theta < 1$ : it is  $\Theta(n^{-\lfloor 1/\theta \rfloor/2})$ , with an order-crossover scale  $n^{1-1/(2(q-1))}$  near each  $\theta = 1/q$  (so the fraction is  $\Theta(1/n)$  at  $\theta = 1/2$ ); amplitudes and crossover functions stay open.

## 75. Throwback Dynamics with Repeated Entries

[enumerative-combinatorics/throwback-queue-dynamics/throwback\\_research.pdf](#)  
from [throwback\\_research.zip](#)

For the queue process that removes the head and reinserts it at the zero-based position equal to its weight: every token recurs if and only if at most  $m$  input tokens have weight at most  $m$ , for every  $m \geq 1$ , with an explicit description of the recurrent tokens when that fails. Adds exact cycle periods and frequencies, a coprimality criterion for perfect mixing, Catalan and prime-parking counts for recurrent cores, and the largest labelled period of a core of size  $r$ .

## 76. Reciprocity and Matrix Duality for Preorder Polytopes

[enumerative-combinatorics/preorder-polytope-reciprocity/article.pdf](#)

from preorder\_reciprocity.zip

Proves Conjectures 7.2, 4.2 and 4.4 of Athanasiadis and Chapoton, *Polytopes and posets associated to preorders* (arXiv:2605.26916v1), and the ordinary  $q = 1$  case of their Conjecture 4.9 (not its full  $q$ -refinement). Two mechanisms: Ehrhart–Macdonald reciprocity after subtracting the all-ones vector, extending the double identity to independently weighted capacities; and duplication of both colour classes of the bipartite graph followed by Postnikov’s lattice-point formula, giving the conjectured matrix transpose.

### 77. Support reciprocity and the $q$ -zeta polynomial of preorder lattice-point posets

[enumerative-combinatorics/preorder-q-zeta-reciprocity/article.pdf](#)

from preorder\_qzeta\_reciprocity.zip + preorder\_q\_zeta\_reciprocity.zip +  
preorder\_q\_reciprocity\_research.zip

Proves the full  $q$ -refinement of Athanasiadis–Chapoton Conjecture 4.9, equation (21): for a preorder  $\tau$  on  $[n]$ , with  $P_\tau$  the nonnegative integer vectors satisfying  $a(I) \leq |I|$  on every order ideal and  $B_\tau$  its maximal elements,  $Z_q(P_\tau, [-1]_q) = (-1)^n \sum_{b \in B_\tau} q^{-|\text{supp}(b)|}$ , together with the stronger identity at each exact support. The  $q = 1$  case was already covered by the preorder-polytope reciprocity report and is cited, not reclaimed. *Merged from three reports that reached the same two theorems by three architecturally independent routes — an incidence inversion with Boolean localization, the same with a Hall-based privatization and a self-contained Euler identity, and a capacity-enumerator reciprocity with a transfer involution — and all three survive.*

### 78. An explicit shelling for preorder lattice-point posets

[enumerative-combinatorics/preorder-shellability/article.pdf](#)

from preorder\_shellability\_research.zip

An affirmative solution to Question 4.6 of Athanasiadis and Chapoton (arXiv:2605.26916v1, p. 13): an explicit pure shelling of the order complex of  $P_\tau$  with its minimum removed, and a coning that shells the order complex of  $P_\tau$  itself. Both complexes are therefore Cohen–Macaulay over the integers and over every field. A different question of that paper from the three other preorder entries.

### 79. Multiple-chain preorder polytopes

[enumerative-combinatorics/multiple-chain-exponential-formula/article.pdf](#)

from multiple\_chain\_exponential\_formula\_research.zip + multiple\_chains\_research.zip

Proves the exponential generating-function identity proposed in Section 9.5 of Athanasiadis and Chapoton (arXiv:2605.26916v1) for lattice points of multiple-chain preorder polytopes counted by number of nonzero coordinates, in a rectangular generalization with independent block width and height. Adds a duality corollary, an Ehrhart evaluation, a degree- $2^d$  root theorem with a quartic elimination certificate, two treatments of the  $\gamma$ -vector, a Stieltjes moment representation, the boundary constant in two forms, and a central limit theorem with exact variance. *Merged; the identity is proved twice, once through a subset encoding with a bounded symbol and once through slack meanders in a Laurent ring, and the eleven deliberate redundancies are listed in the text.*

### 80. Rationality and growth of stabilized permutation-weight series

[enumerative-combinatorics/stabilized-permutation-weights/article.pdf](#)

from stabilized\_permutation\_weights.zip

Studies the full stabilized generating functions of the permutation-weight  $q$ -Eulerian polynomials, beyond their initial connection with OEIS A256193: an all- $d$  finite-spectrum theorem, rationality, a universal integer denominator, the exact radius of convergence and residue-class asymptotics. The two-descent sequence gets an explicit formula in powers of 3, powers of 2 and Fibonacci numbers.

### 81. A Reflexive Root-Polytope Model for Preorder $h$ -Polynomials

[enumerative-combinatorics/preorder-root-polytopes/article.pdf](#)

from preorder\_root\_polytopes.zip; batch 39: ProveIt\_Height\_Two\_Preorder\_Research.zip; batch 42: ProveIt\_Cactus\_Rigidity.zip; batch 44: ProveIt\_Preorder\_Gamma.zip,  
ProveIt\_Preorder\_Gamma\_Research.zip; batch 55: ProveIt\_Lorentzian\_Support.zip,

ProveIt\_Support\_Polynomials\_Research.zip, ProveIt\_ADE\_Stability.zip, ProveIt\_Matroid\_Lifts.zip;  
batch 63: ProveIt\_Marked\_Tree\_Stability.zip; batch 64: ProveIt\_Two\_Flow\_Lattice\_Sampling.zip;  
batch 66: ProveIt\_Random\_Tree\_Stability.zip; batch 68: ProveIt\_Hall\_Bottlenecks.zip

Proves Conjecture 5.1(c) of Athanasiadis and Chapoton (arXiv:2605.26916v1): for a preorder  $\tau$  on  $[n]$  the directed root polytope  $C_\tau = \text{conv}(\{\pm e_i\} \cup \{e_i - e_j : j \leq_\tau i\})$  is reflexive with  $h^*(C_\tau, t) = h_\tau(t)$ , and any pulling triangulation of its boundary is combinatorially the boundary of a simplicial polytope with that  $h$ -polynomial. Palindromicity and unimodality, Conjecture 5.1(a,b), follow, as do the full  $g$ -theorem conditions. The higher-dimensional augmented bipartite identity and preorder duality are credited to Dai, Hou, Liu, Thawinrak and Wang (arXiv:2608.16037v2). Part II, added in batch 39, treats height-two preorders through matching polynomials:  $\gamma$ -positivity in height two (Conjecture 5.2 there), a family showing that real-rootedness (Conjecture 5.3) fails — its first counterexample is credited to a MathDB posting by Shivam Patel — and crown obstructions: an induced crown forces large minimal nonfaces in every lattice triangulation of  $C_\tau$ , so the boundary complex is not flag when the crown has at least six elements. Flag realizability in general (Conjecture 5.1(d)) remains open. Part III, added in batch 42, shows that exactly the cacti admit a faithful unit-phase minor model, a property strictly stronger than real stability. Part IV, merged in batch 44 from two manuscripts, proves Conjecture 5.2 for every finite preorder:  $\gamma_k$  counts ordered pairs of disjoint  $k$ -sets joined by a  $\tau$ -increasing bijection, and a reflexive relation gives a palindromic polynomial exactly when it is transitive. Part V, merged in batch 55 from two manuscripts, shows that for a bipartite graph with shores of sizes  $m$  and  $n$  every support enumerator divided by  $\binom{\min(m,n)}{k}$  is log-concave (the unweighted inequality is Theorem A of Röhrle and Ulirsch, arXiv:2402.15317, carried through the support-counting bridge), so every  $h_\tau$  and every height-two  $\gamma$ -vector is ultra-log-concave; it classifies the palindromic support polynomials of all bipartite graphs, gives FPRASes for partition functions and for single coefficients, and shows exact counting  $\#P$ -complete already in height two. Part VI, added in batch 55 with the stability sections of a further manuscript, shows that stability of the signed support polynomial is block-local, holds for the incidence–apex graph of a tree exactly when the tree is a finite or affine ADE tree, and is classified for books and bipartite multi-theta graphs; the books show it strictly stronger than real-rootedness. Part VII, added in batch 55, shows that the transversal matroid of a preorder determines the preorder up to isomorphism and that height-two preorder matroids generate exactly the gammoids under minors. Part VIII, added in batch 63, classifies stability for one apex adjacent to an arbitrary set of vertices of a tree (a corollary of Part VI, also proved directly), with exact  $O(n^5)$  enumeration, optimization and repair of the stable apex neighbourhoods, a sharp diameter window, and real-rootedness of their size-counting polynomial exactly for paths. Part IX, added in batch 64, gives explicit polynomial-time two-flow bijections between the demand vectors of a bipartite graph with prescribed support and the supplier sets matchable to it (the correspondence itself is Oh’s), and with them an almost-uniform sampler of actual lattice points of every finite preorder polytope. Part X, added in batch 66, answers Part VIII’s question for a uniform random labelled tree with every vertex marked independently with probability  $p$ : the probability that the marking is stable is asymptotic to  $\frac{9}{8}\sqrt{2\pi} p^3 n^{5/2} e^{-pn}$  whenever  $p\sqrt{n} \rightarrow \infty$ , an explicit entire function governs the window  $p\sqrt{n} \rightarrow \lambda$ , and below it stability is asymptotically the event of at most two marks; for a fixed host the diameter determines the probability up to a polynomial factor, separating the typical rate  $\log(1-p)$  from the annealed rate  $-p$ . Part XI, added in batch 68, answers Part V’s question on normalization by the matching number negatively for positive vertex weights on both shores: a graph with 30 vertices, 104 edges, matching number eight and weights in  $\{1, 500\}$  has a weighted support polynomial that is not ultra-log-concave of order eight. A Hall-bottleneck family gives a rank-six phase diagram, the endpoint threshold for every core size and a lower bound asymptotic to  $\frac{4}{3}\nu$  for any normalization depending on the matching number  $\nu$  alone; the first rank-normalized inequality holds for every positively weighted graph, with a defect identity. The unit-weight and one-shore versions stay open. Conjectures 7.2, 4.2 and 4.4 of the same paper are proved in the preorder-polytope reciprocity report.

## 82. Largest Simplex Faces under Ordinal Sums

enumerative-combinatorics/

simplex-faces-of-order-and-chain-polytopes/Largest\_Simplex\_Faces.pdf

from Series\_Parallel\_Simplex\_Faces\_Research.zip

Answers the concluding open question of Freij-Hollanti and Lundström, *Simplex inequalities of order and chain polytopes of recursively defined posets* (Order 43 (2026), article 31). For  $s_O(P)$  and  $s_C(P)$ , the largest dimensions of simplex *faces* of the order and chain polytopes: both are computable in  $O(n)$  on any series-parallel decomposition, the order case carrying eight normalized endpoint states. For an ordinal sum of antichains,  $s_O = r + \nu$  with  $\nu$  a maximum  $b$ -matching of the layer path under capacities  $\min(a_i - 1, 2)$ , and  $s_C = r + \max(0, k - 1)$ . The extremal gap is sharp:  $0 \leq s_C - s_O \leq \lfloor (n - 2)/3 \rfloor$ , attained for every  $n \geq 2$ .

## 83. Skew partitions, continued fractions and connected components

enumerative-combinatorics/skew-partition-continued-fractions/article.pdf

from A225114\_proof\_bundle.zip

Proves the repeated-power continued-fraction formula recorded as conjectural in OEIS A225114, which counts skew partition diagrams with no empty row or column. Unique concatenation gives  $A(q) = 1/(1 - P(q))$  for the edge-connected diagrams, and a boundary-path encoding followed by a weighted two-coloured Motzkin-path decomposition gives  $P(q) = q/(1 - 2q - q^3/(1 - 2q^2 - q^5/(1 - 2q^3 - \dots)))$ , whence  $A(q) = 1/(2 - C(q))$  with  $C = 1 + P$ . Adds a four-variable refinement, a formula by height, width and component count, a sharp truncation bound, exponential asymptotics and a central limit theorem for the number of components. **Explicitly not a novelty claim:** a prior-art check found Pasechnik’s MathOverflow sketch of ten September 2024, and the entry credits Kurkov; the report says so first and credits both.

## 84. From mesh avoidance to Catalan inflation

enumerative-combinatorics/mesh-avoidance-catalan-inflation/article.pdf

from A289587\_proof.zip, OEIS\_A289587\_proof.zip

Proves Scheuerle’s generating-function conjecture in OEIS A289587 for permutations avoiding 321 and the mesh pattern (12, 174), and the equal count for (12, 234) by inversion: with  $D = x^4 - 2x^3 - 5x^2 - 2x + 1$ ,  $P = x^4 + 4x^3 + 11x^2 + 10x + 3$  and  $Q = x^2 + 5x + 3$ , the generating function is  $(P - Q\sqrt{D})/(8x(1 + x)^2)$ . The headline structural result is the exact *occurrence count*  $\text{occ} = \binom{f}{2} + \sum_U \binom{|U|}{2}$ , of which avoidance is the case  $\text{occ} = 0$ . Merged from two reports that reached the identical generating function by the identical run-contraction substitution  $u \mapsto u/(1 + xu)$ ; the merge keeps *both* refinements, because they refine the same sequence by different statistics — excedances and records — and are genuinely different theorems, agreeing only in total. The two local predicates they are built on disagree on 31 of the 321-avoiders of length at most six, the smallest being 12, and the merged article proves where they coincide rather than assuming it.

## 85. Prescribed coefficient-lattice minima

enumerative-combinatorics/coefficient-lattice-minima/article.pdf

from coefficient\_lattice\_minima\_research.zip

Settles the unnumbered realization conjecture after Lemma 1 of O’Bryant, *On Nathanson’s Triangular Number Phenomenon* (arXiv:2506.20836): every nondecreasing list of even integers is the list of successive  $L^1$ -minima of the coefficient lattice of some finite integer set. The witness is the mixed-radix set  $A = \{0, 1, h_1, h_1 h_2, \dots\}$ , whose minima are exactly  $(2h_1, \dots, 2h_r)$  with  $r + 2$  distinct elements, and the radices may be given in any order.

## 86. Coefficientwise magic positivity through dimension ten

enumerative-combinatorics/magic-positivity-parking-polytopes/article.pdf

from magic\_positivity\_research.zip

A polytope is *magic positive* when its Ehrhart polynomial has nonnegative coefficients in the basis  $t^j(t + 1)^{n-j}$ . Proves, in exact arithmetic, that the generalized parking-function polytope  $X_n(b)$  is magic positive for every dimension  $3 \leq n \leq 10$  and *every* positive integer parameter vector, uniformly in  $b$  rather than on a sampled box — a partial solution of Conjecture 8.1 of Hill, Luo,

Trinh and Vindas-Meléndez (arXiv:2607.15503v1). Dimension three is proved by printed algebraic formulas; the first-coefficient harmonic-number formula and the endpoint identities hold in every dimension. The conjecture in unrestricted dimension remains open.

### 87. Monotone blocks and reverse replies

enumerative-combinatorics/reverse-reply-avoidance-game/article.pdf  
from reverse\_reply\_solution.zip

Proves Conjecture 2.9 of Ulfarsson (arXiv:2603.16004): for every  $k \geq 3$  and every  $n \geq R(k) = k^4 - 2k^3 - 3k^2 + 5k + 5$ , the second player wins the normal-play permutation-avoidance game on  $S_n$  with pattern length  $k$  by always replying with the reversal, so the Sprague–Grundy value is 0. The proof is elementary, with no computer search: a minimal-support lemma isolates one pattern inside a family built from singletons plus one inflatable monotone interval, and inflation restores the original length so the reply is legal.  $R(k)$  is asserted sufficient, not minimal.

### 88. Nonintegral tropical degree from two horizontal segments

enumerative-combinatorics/tropical-hilbert-nonintegral-degree/article.pdf  
from tropical\_hilbert\_counterexample.zip

Computes the strict-witness tropical Hilbert function of Grigoriev (arXiv:2404.06440v2) exactly for a two-segment set:  $\text{TH}_{V_d}(k) = k + \min\{k, \lceil k/d \rceil\} + 1$ . Taking  $d = 3$  gives tropical degree  $4/3$  — not an integer — and a quasipolynomial of least period 3, so eventual polynomiality fails. Every rational in  $(1, 2]$  is realized, denominators are unbounded, and for  $d = p/q > 1$  in lowest terms the least eventual period is exactly  $p$ . Both bounds are proved by hand.

### 89. Finite-prefix corrections for polyomino growth

enumerative-combinatorics/polyomino-growth-finite-prefix-corrections/article.pdf  
from ProveIt\_Polyomino\_Research.zip; batch 40: ProveIt\_Spectral\_Saturation.zip

Continues the formal project Combinatorics/Polyominoes/KlarnerConstant, whose Lean-verified bound is  $\lambda \leq 9047/2000 = 4.5235$  for Klarner’s constant. Proves a finite-prefix correction theorem for proper positive polynomial recurrences and applies it to Bui’s seventeen-neighbourhood system with an exact marked-occurrence table through size 18, giving the computer-assisted, unformalized bound  $\lambda \leq 2249/500 = 4.498$  and  $A_n \leq (927884613/10^9)(2249/500)^n$ . A second theorem gives balanced integer-monomial divergence certificates, complete for productive, strongly connected, nonlinear systems, and yields  $4.52349 \leq \mu_{\text{original}} \leq 4.5235$  for the original recurrence — a statement about the recurrence, not a lower bound on  $\lambda$ . At intake, a fresh build of the delivered enumerator (the same program, not an independent one) regenerated the size-18 table and matched it exactly. Part II, added in batch 40, shows where the method stops: exact spectral saturation in the irreducible critical case, a square-root law, and an all-prefix barrier  $\mu_N \geq 4.3149$  for Bui’s map — a limit of this method, not a lower bound on  $\lambda$  and not an improvement on 4.498.

## 12 Generating functions and asymptotics

### 12.1 Coefficient counts of binary products

Three reports on counting coefficients of  $\prod_i (1 + x^{G_i})$ . The first two answer, independently, the universal-rationality part of Richard Stanley’s MathOverflow question 431075 (“A conjectured rational generating function”, 2022), which had no posted answer when checked on 19 September 2026. A third answer used the same mixed-base construction and the same enumerative argument as the first, and has been merged into it.

### 90. A Mixed-Base Counterexample to Stanley’s Rationality Question

generating-functions-and-asymptotics/binary-product-coefficient-counts/  
mixed-base-counterexample/article.pdf  
from stanley\_mixed\_base\_counterexample.zip + Stanley\_Rationality\_Counterexample.zip

With  $G_{2j+1} = 2^j$  and  $G_{2j+2} = 3^j$  the exponent generating function is rational, while  $\nu(2n) = 2^{n - \lfloor (n+1)\alpha \rfloor}$  for  $\alpha = \log_3 2$  counts the coefficients equal to one in  $\prod_{i \leq N} (1 + x^{G_i})$  and its generating

function is not rational, not D-finite, not P-recursive, and has a natural boundary. Classifies rationality in the family interleaving  $a^j$  and  $q^j$  with digit polynomial  $1 + y + \dots + y^{a-1}$  for all  $2 \leq a < q$  — Stanley’s general  $r = a - 1$  — as holding exactly when  $q^u = a^v$ . Adds a finite-quotient rigidity theorem, an abstract natural-boundary theorem for  $\sum q^{\lfloor n\theta \rfloor} t^n$  at every irrational  $\theta$ , the exact limit set of  $a_n/\lambda^n$ , and a streaming recurrence making the point that non-P-recursive does not mean hard to compute. *Merged; the enumerative half was the same argument in both originals, but the two routes to non-rationality differ and both are kept — an elementary finite-state reduction mod  $p$ , and the rigidity route that reaches the stronger transcendence conclusions.*

### 91. Rational exponents, nonrational coefficient counts

[generating-functions-and-asymptotics/binary-product-coefficient-counts/recursive-exponent-construction/research.pdf](#)  
 from stanley\_counterexample.zip

A different negative answer, from the recursion  $a_0 = 2$ ,  $a_1 = 1$ ,  $a_n = a_{n-1} - 2a_{n-2}$ , with  $b_n = T_{n-1} + 2^n - 1 + a_n$ ,  $T_n = T_{n-1} + b_n$ ,  $G_{2n-1} = 2^{n-1}$  and  $G_{2n} = b_n$ : the counting series has a natural boundary on the unit circle, hence is neither rational, algebraic nor D-finite.

### 92. Coefficient residues in generalized Fibonacci products

[generating-functions-and-asymptotics/binary-product-coefficient-counts/kbonacci-coefficient-residues/article.pdf](#)  
 from stanley\_kbonacci\_research.zip + quasifibonacci\_parity\_research.zip

Settles Conjectures 6.2 and 6.3 of Stanley, *Theorems and Conjectures on Some Rational Generating Functions* (arXiv:2101.02131v3, pp. 23–24). For  $k$ -bonacci seeds each exceeding the sum of their predecessors, the number of odd coefficients of  $\prod_{i \leq n} (1 + x^{w_i})$  has generating function  $(1 + 2z^k)/(1 - 2z + 2z^k - 2z^{k+1})$ , independently of the seeds; the proof builds periodic signs of block length  $k + 1$  and block product  $-1$  and shows every signed finite product is flat. *Merged; the Conjecture 6.3 halves proved the same statement on the same domain, and the surviving report adds Conjecture 6.2 and the asymptotic analysis.*

## 12.2 Single-sequence OEIS studies

Nine reports on the growth or the closed form of a single OEIS sequence. Three of them count the terms in the derivatives of a power tower — of  $x^{x^x}$ ,  $x^{x^2}$  and  $x^x$  — and reach three different answers by three different recurrences; two more share a Bell-scale saddle-point method on unrelated sequences. Shared technique is not shared theorem, and none of these nine duplicates another.

### 93. Bell-scale growth of OEIS A088714

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a088714-bell-scale-growth/article.pdf](#)  
 from A088714\_asymptotic\_growth.zip

For the unique formal series with  $A(x) = 1 + xA(x)^2A(xA(x))$ , proves  $\log a_n = n \log n - n \log \log n - n + O(n \log \log n / \log n)$ , so  $(e \log n / n) a_n^{1/n} \rightarrow 1$  and the ordinary generating function has radius of convergence zero while the exponential one is entire. The lower bound comes from a single explicit path through an exact triangular recurrence, the upper from a Stirling-positivity comparison with derivatives of Touchard polynomials. The multiplicative equivalent is *not* claimed, and the finer Bell-number form is labelled a conjecture with a sufficient ratio condition isolated.

### 94. The exponential growth constant of OEIS A158415

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a158415-growth-constant/article.pdf](#)  
 from A158415\_asymptotic\_growth.zip

For the number of distinct values expressible with  $n$  symbols from  $E ::= 1 \mid \sqrt{E} \mid (E + E)$ , proves a computable  $\gamma > 1$  exists with  $a(n)^{1/n} \rightarrow \gamma$  and  $1.86204 < \gamma < 1.9208$ , gives an effective characterisation as a supremum of algebraic finite-data bounds, and proves the boundary identity  $\sum a(n)\gamma^{-n} = \gamma^2$ , which rules out a pure  $C\gamma^n$  prefactor. The rigorous lower bounds rest on a

uniform injectivity theorem: any finite alphabet of representable values can serve as nested-radical digits, via an odd valuation at a prime above 2. No closed form for  $\gamma$  is claimed, and terms 22–28 get certified intervals rather than exact values.

### 95. Dyadic scaling rigidity of 3AP-free permutations

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a003407-dyadic-scaling-rigidity/article.pdf](#)

from ProveIt\_Dyadic\_Scaling\_Rigidity.zip (batch 40); batch 43: ProveIt\_3AP\_Nonholonomicity.zip

For  $\theta(n)$ , the number of permutations of  $[n]$  with no three-term arithmetic progression (OEIS A003407), studies the dyadic scaling of  $\theta(n)$ : a continuous log-periodic growth profile with its exact fundamental period, rigidity of the profile under real rescaling, explicit losses and gains of the 2:1 gluing, and the empirical and sampling laws; the adjacent ratios  $\theta(n+1)/\theta(n)$  have no limit. The parity recurrences and Sharma’s Theorem 2.8, which the manuscript imports, are already proved in Lean in `Combinatorics/Ramsey` (the table  $\theta(1..20)$  there rests on a `native_decide` certificate); the report’s own theorems are not formalized. Part II, added in batch 43, proves  $\theta$  is not P-recursive and its generating functions are not D-finite (the ordinary one is transcendental), using the published counts  $\theta(64)$ ,  $\theta(75)$ , which nothing in ProveIt certifies; whether the generating function has a natural boundary stays open.

### 96. An exact coefficient formula for Ramanujan’s theta function

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a260306-formula-proof/a260306\\_proof.pdf](#)

from A260306\_formula\_proof.zip

Proves both finite formulas recorded in OEIS A260306, for every coefficient index, for the expansion  $\theta(N) \sim \sum c_n N^{-n}$  defined by  $\sum_{k < N} N^k/k! + \theta(N)N^N/N! = e^N/2$ . The route is an exact Laplace-integral difference, an analytic change of variable with  $F(\psi(w)) = w^2/2$ , and Lagrange inversion reducing each  $c_n$  to one coefficient extraction, then a finite change of basis into the entry’s double and triple sums. Includes the analytic error estimate the identification needs, rather than only checking the two sums agree, and identifies A065973 as the reduced denominators.

### 97. The maximal positive coefficient of a Hermite polynomial

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a277280-hermite-maximum/article.pdf](#)

from A277280\_asymptotics.zip

An exact closed form for OEIS A277280, the largest coefficient of the physicists’ Hermite polynomial  $H_n$  with signs retained: with  $k_n = \lfloor (2n+6 - \lfloor \sqrt{8n+21} \rfloor)/8 \rfloor$  and  $d_n = n - 4k_n$ , the exponent  $d_n$  is the unique maximizer and  $a(n) = n! 2^{d_n} / ((2k_n)! d_n!)$ . The distinction from the largest coefficient in absolute value is kept explicit throughout.

### 98. Asymptotics of OEIS A277364

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a277364-bell-asymptotics/A277364\\_asymptotics.pdf](#)

from A277364\_asymptotics.zip

For  $a(n) = \sum_{k \leq n/2} S(n, k)$ , partitions of an  $n$ -set into at most  $\lfloor n/2 \rfloor$  blocks, proves  $a(n) \sim B_n$  with an explicit first correction: writing  $r = W_0(n)$ ,  $a(n) = \exp\{n(r-1+1/r)-1\}(1+r)^{-1/2}(1-P(r)/n+O(r^2/n^2))$  for an explicit rational  $P$ . Also gives a parity-split equivalent for the omitted tail  $B_n - a(n)$ .

### 99. Term counts in the derivatives of power towers

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/power-tower-derivative-term-counts/article.pdf](#)

from A293239\_research\_report.zip, A290268\_partial\_results.zip, A281434\_exact\_algorithm\_and\_cubic\_growth.zip; batch 42: ProveIt\_A290268\_Finite\_Certificates.zip

One article on three OEIS sequences that ask the same question of three functions: how many distinct monomials are in the fully collected  $n$ th derivative of  $x^x$  (A293239),  $x^{x^2}$  (A290268) and  $x^{x^x}$  (A281434)? Part I proves once the apparatus all three had established separately —

the canonical differential-polynomial form  $f^{(n)} = fP_n$  or  $fx^{-n}P_n$  with its first-order integer recurrence, the independence of logarithmic monomials that makes the count an invariant of  $f$  rather than of a differentiation algorithm, the identity  $a(n) = |\text{supp } P_n|$ , and the envelope bound. Two normalizations are needed, because neither clause covers all three in the frames the parts use. **None of the three headline questions is fully settled, and the article says so in its abstract:** for  $x^x$  the count reduces exactly to zeros of the first-kind Lehmer–Comtet triangle A008296 and  $a(n) = \Theta(n^2)$ , but the closed formula needs an isolated nonvanishing step — and separately the recurrence printed in the entry for  $n > 6$  is *disproved* at  $n = 8$ ; for  $x^{x^2}$  the conjectured expression is proved to be an upper bound and  $a(n) = \Theta(n^2)$ , but not the constant  $\frac{1}{2}$ ; for  $x^{x^x}$  explicit bounds give  $a(n) = \Theta(n^3)$  but not  $a(n) \sim \frac{2}{3}n^3$ , and there the obstruction is not a nonvanishing statement but a density one, since further zeros provably exist. A closing part sets the three side by side and records what they share beyond the framework: every cancellation any of them can prove is a parity zero of the same falling factorial. This is a thematic consolidation, not a deduplication: the three were checked and are not duplicates, and all fifty numbered results of the three originals are preserved unchanged. Batch 42 added to the  $x^{(x^2)}$  part depth certificates that close the nonvanishing problem at logarithmic deficits 3 and 4 for every  $n$  and improve its lower bound on the count; the open region is now deficit at least 5. Where that manuscript re-derived results already in the report, it is printed as a second route.

#### 100. Iterated sums and products from one

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a352969-correctness-and-growth/article.pdf](#)  
from A352969\_correctness\_and\_growth.zip

For  $S_0 = \{1\}$ ,  $S_{n+1} = (S_n + S_n) \cup (S_n \cdot S_n)$  and  $a_n = |S_n|$  (OEIS A352969), proves correctness of the listed terms and the doubly exponential law  $2^{2^{n-rn-4}} \leq a_n \leq 2^{2^{n-1}}$ , hence  $\log \log a_n = n + O(\sqrt{\log n})$ . The step is simultaneous, not repeated closure — a distinction the report is careful about.

#### 101. Distinct third derivatives of a power tower

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a199085-exponentiation-derivative-values/article.pdf](#)  
from A199085\_generating\_function.zip

For OEIS A199085, the number of distinct values taken by the third derivative at  $x = 1$  over all fully parenthesized exponentiations of  $n$  copies of  $x$ , proves  $a(1) = a(2) = 1$  and  $a(n) = \lfloor (n^2 - 2)/3 \rfloor$  for  $n \geq 3$ , with the rational generating function  $z/(1-z) + z^3/(1-z)^3 - z^7/((1-z)^2(1-z^3))$ . The proof is for all  $n$ , not an extrapolation: every attainable pair of second and third derivatives is classified and a witness constructed for each admissible pair. Despite the shared vocabulary this is a different question from *power-tower-derivative-term-counts*, which fixes the function and varies the derivative order; here the order is fixed at three and the parenthesization varies.

### 12.3 Asymptotics, records and discrepancy

#### 102. The Exponential Scale of Gregory Sign Thresholds

[generating-functions-and-asymptotics/gregory-coefficient-sign-thresholds/gregory\\_sign\\_threshold.pdf](#)  
from gregory\_sign\_threshold\_research.zip

For  $(x/\log(1+x))^m = \sum_n G_n^{(m)} x^n$  and  $\text{Gr}(m)$  the least  $N$  beyond which  $(-1)^{n-1} G_n^{(m)} > 0$ , proves  $\log \text{Gr}(m) = m + 1 - \gamma - (\zeta(2) + \zeta(3))/m + C_2/m^2 + O(m^{-3})$  with  $C_2$  explicit, plus a finite algorithm for every coefficient of the complete expansion; in particular  $\text{Gr}(m)/e^m \rightarrow e^{1-\gamma}$  and  $\text{Gr}(m)/3^m \rightarrow 0$ . This replaces Conjecture 6.1 of Xu and Zhao (arXiv:2609.11072v1), whose printed lower bound already conflicts with its own reported  $\text{Gr}(5) = 113$ ; that value is independently certified here.

#### 103. Exact Records in the Square-Root-of-Three Walk

[generating-functions-and-asymptotics/sqrt-three-walk-records/article.pdf](#)  
 from [sqrt3\\_record\\_recurrence.zip](#)

Proves the four-phase record recurrence for  $S(N) = \sum_{n \leq N} (-1)^{\lfloor n\sqrt{3} \rfloor}$  in the corrected form on page 7 of Bruin and Fokkink, *On the records and zeros of a deterministic random walk* (arXiv:2503.11734v2), where it is experimental and attributed to Arias de Reyna and van de Lune. Includes minimality of every claimed first passage, a rational generating function, closed forms, sharp logarithmic extrema, and an extension to every even integer trace  $D \geq 4$ .

#### 104. Sharp discrepancy for a family of binary substitutions

[generating-functions-and-asymptotics/binary-substitution-discrepancy/article.pdf](#)  
 from [oeis\\_discrepancy\\_counterexamples.zip](#)

For the fixed word of  $0 \mapsto 1, 1 \mapsto 101010$ , the position sequences A284365 and A284366 both carry a conjectured upper position error below 2, and both conjectures are false: the first counterexamples are at ranks 2,977,771 and 8,933,313, with common error  $(2977771\sqrt{21} - 13645857)/2 \approx 2.00487$  and sharp common upper bound  $(6 + \sqrt{21})/5 \approx 2.11652$ . Proves complete discrepancy limit intervals for every substitution  $0 \mapsto 1, 1 \mapsto (10)^m$ , with infinite counterexample families and exact recurrences for their indices.

#### 105. Exact eventual maximizers of Rudin–Shapiro autocorrelations

[generating-functions-and-asymptotics/rudin-shapiro-autocorrelation-maximizers/article.pdf](#)  
 from [rudin\\_shapiro\\_maximizers.zip](#)

For the aperiodic autocorrelation  $C_m(k)$  of the level- $m$  Rudin–Shapiro sequence, classifies the maximizing shift exactly for all large  $m$ :  $k_m^* = (2^{m+1} + (-1)^m)/3$ , with  $A_m = b_{m-2}$ , settling the discrete uniqueness-and-location conjecture of Tarnu (J. Approx. Theory 287 (2023) 105866), repeated as Conjecture 1.5 of Choi–Tarnu (2025), including the limit  $k_m^*/2^m \rightarrow 2/3$ . The certificate is a rational polytope argument with exact arithmetic.

#### 106. Apéry arrays and conjectural zeta accelerations

[generating-functions-and-asymptotics/apery-array-zeta-accelerations/article.pdf](#)  
 from [Apery\\_OEIS\\_Conjectures\\_Proofs.zip](#), [apery\\_oeis\\_conjectures.zip](#)

Proves the entire shifted-diagonal  $\zeta(3)$  family recorded as conjectural in OEIS A143007, and both the super- and subdiagonal  $\zeta(2)$  families of A108625, at every nonnegative offset: for all  $k \geq 0$ ,  $\zeta(3) = H_k^{(3)} + \sum_{n \geq 1} (2n+k)(3n^2+3nk+k^2)/(n^3(n+k)^3 T_{n-1,n+k-1} T_{n,n+k})$ . The main diagonals are the classical Apéry sequences A005259 and A005258. Proofs are finite telescoping identities with rational lattice potentials and explicit limit estimates; no computer test or unproved conjecture enters. Adds rational error bounds, numerator and denominator recurrences, a three-term recurrence along every shifted diagonal, and sharp fixed-offset asymptotic constants. Merged from two reports whose boxed identities were character-identical up to macro names.

### 13 The Jacobian conjecture and Keller maps

Three reports continuing the formal project [Algebra/JacobianConjecture](#), which verifies in Lean and Rocq Alpöge’s three-variable counterexample and a degree-six stable representative in five variables (its stabilization to all higher dimensions in Lean only); its further degree reductions are SymPy certificates. None of the three reports’ own theorems is formalized; each README names the project declarations it builds on. Two study the project’s three-variable map; the third studies a different map, Gao’s five-dimensional  $F_6$ .

#### 107. The fiber geometry of Gao’s Keller map $F_6$

[jacobian-conjecture/gao-f6-fiber-geometry/article.pdf](#)  
 from [ProveIt\\_Six\\_Sheets\\_Research.zip](#), [ProveIt\\_Keller\\_Fibers\\_Research.zip](#)

For Gao’s five-dimensional six-sheeted Keller map  $F_6$  (arXiv:2608.00222v1, §4.5.1), classifies every geometric fiber: the cardinalities are exactly 0, 1, 2, 3, 4, 6, all seven root-multiplicity patterns

occur, and off the hyperplane  $c = 0$  the inverse points are the simple roots of an explicit sextic. The omitted set is a smooth closed surface isomorphic to  $\mathbb{G}_m \times \mathbb{A}^1$ , of codimension three, with the single effective omitted parameter  $(21/32, 5/32, 25/128)$ . Gives an explicit finite normalization of rank six, the two-component nonproperness locus, the discriminant's normalization and local analytic types, global monodromy  $S_6$  with two arithmetic certificates, no proper intermediate field, real fiber cardinalities  $\{0, 1, 2, 3, 4\}$ , and an obstruction to equivalence with a stabilization of a map in at most three variables. The map is Gao's; the fiber theory is the reports'. Merged from two manuscripts proving the same core in identical normalizations; the second one's different proofs are kept as marked second routes. Not to be confused with the project's formally verified dimension-five stable map.

### 108. Arithmetic local–global dichotomies for the three-variable Keller map

[jacobian-conjecture/arithmetic-local-global-fibers/article.pdf](#)

from ProveIt\_Arithmetic\_Local\_Global.zip; batch 58: ProveIt\_Dense\_Integral\_Hasse.zip; batch 64: ProveIt\_Beyond\_Split\_Fibers.zip

For the project's three-variable Keller map, gives an explicit infinite family of integral targets with a preimage in every  $\mathbb{Z}_p^3$  but none in  $\mathbb{Z}^3$  (each fiber is three rational points, so these are not rational Hasse failures); classifies the split integer-root plane  $C = 2$  by a gcd condition, with locally soluble parameter density  $27/(4\pi^2)$  surviving the removal of  $S$ -integral fibers; proves the exact integral 2-adic fiber law  $(21, 7, 3, 1)/32$  for fiber sizes 0, 1, 2, 3, determined modulo 8; and contrasts it with a rational Hasse principle over every number field. The complex stratification is credited to Gao and the finite-field histograms to an external note. An intake remark records an integral triple collision over the project's own collision value. Not a new disproof of the Jacobian conjecture. Part II, added in batch 58, answers Part I's density question: such failures are Zariski dense in every plane  $C = c \neq 0$ , hence in  $\mathbb{A}^3$ , and none lies on  $C = 0$ . It gives a complete local criterion for completely split fibers, finds finitely many integral split fibers on each plane, and proves  $\#\mathcal{H}_c^{\text{sp}}(T) \sim \kappa(c)T^{2/3}$  for the completely split failures in square boxes  $|A|, |B| \leq T$ , with  $\kappa(c)$  an explicit Gamma factor times an Euler product. Part III, added in batch 64, gives a complete local-integral certificate for every target, nonsplit fibers included, and on  $C = 2$  counts the integral Hasse failures in the same boxes:  $\frac{27}{2\pi^2}T \log T + O(T)$ , almost all of them nonsplit.

### 109. All-degree rigidity of a weighted Keller class

[jacobian-conjecture/weighted-keller-rigidity/article.pdf](#)

from ProveIt\_Weighted\_Keller\_Rigidity.zip; batch 44:

ProveIt\_Affine\_Weighted\_Keller\_Classification.zip

For polynomial maps of weights  $(-1, 1, 2)$  written through  $t = xy$  and  $v = x^2z$ , classifies the Keller maps in every degree over any field of characteristic zero: those with  $p, q$  affine in  $v$  and  $r$  affine in  $(t, v)$  are an explicit tame family and a two-parameter diagonal orbit of the known map, so every noninvertible member has coordinate degrees  $(7, 6, 4)$  and sixteen monomials, with no input degree bound. Extends to  $r = R(t) + \gamma v$ , where the possible maximum degrees of the noninvertible members are exactly 7 and every even integer at least 8, and shows that with  $r = 1 + t + v$  the fixed degree-seven coefficient scheme is the double point  $\mathbb{Q}[\varepsilon]/(\varepsilon^2)$ , with an explicitly obstructed tangent (a torus times it when the two coefficients of  $r$  are freed). The classification does not cover unrestricted weighted maps. Sharpens to all degrees the project's degree-seven sparsity search; it does not show degree seven minimal in dimension three. Part II, added in batch 44, removes the restriction on the coefficient of  $v$  in  $r$ : whenever  $p, q, r$  are affine in  $v$  that coefficient is constant, so the classification covers the whole affine-in- $z$  class; it adds dual-number slopes of every degree and a universal second-order obstruction.

## 14 Galois theory and radicals

Two reports continuing the formal project `Algebra/PolynomialFormulas` (Lean and Rocq root formulas through degree four, Abel–Ruffini, and Lazard's quintic solver); none of its theorems is formalized.

### 110. Specialization-safe radical solvers

[galois-theory-and-radicals/specialization-safe-radical-solvers/article.pdf](#)

from ProveIt\_Radical\_Solvers\_Research.zip; batch 52: ProveIt\_Optimal\_Kummer\_Atlases.zip

Gives an exact cyclotomic-resultant criterion for the characteristics in which equal-cardinality subset sums of the roots of an irreducible prime-degree polynomial can collide, and evaluates it: the bad characteristics are  $\{5\}$  for quintic pairs,  $\{2, 7\}$  for septic pairs and  $\{2, 7, 29\}$  for septic triples. Classifies the collisions completely in characteristic 29 (exactly the cyclic polynomials  $(X - \eta)^7 - b$ ) and in characteristic 2 (translated  $Z^7 + aZ^3 + bZ + d$ , realizing the nonsolvable  $\text{GL}_3(\mathbb{F}_2)$ ). Constructs separating subset invariants in every characteristic, reconstructs all roots from one coherent  $p$ -th radical on each Fourier chart once the intermediate-field invariant data are given (generalizing Lazard’s pivot-one scheme for quintics; a coefficient-only solver is left as further work), realizes every Fourier support by rational polynomials with full affine Galois group, and shows the Picard group of the cyclic configuration quotient is  $\mathbb{Z}/p$ , so no single global Kummer coordinate exists. The Picard result has a written proof only. Part II, added in batch 52, answers Part I’s question on the least number of charts:  $\text{CH}^*(\text{Conf}_p(\mathbb{A}^d)/C_p) = \mathbb{Z}[t]/(pt, t^{d(p-1)})$ , so every cover by regular Kummer charts, even with nonlinear character sections and different characters on different charts, needs exactly  $d(p-1)$  charts ( $p-1$  for polynomials: four for quintics, six for septics), and the Fourier charts attain it. This too is a written proof, by integral intersection theory; its checker verifies only finite identities.

### 111. Finite separating ranges for sextic block resolvents

[galois-theory-and-radicals/sextic-block-resolvent-separators/article.pdf](#)

from ProveIt\_Finite\_Sextic\_Separators.zip (batch 41); batch 57:

ProveIt\_Three\_Plus\_One\_Resolvents.zip

The project finds parameters separating the sextic pair and triple block-resolvent descriptors by an unbounded `Nat.find` search. This report proves in writing that on the curve  $(t, 2t^2)$  one of  $t = 1, \dots, 196$  always separates both families, sharply for arbitrary complex test sets, by orbit counting. The bound is not formalized; formalizing it would make that search bounded, and two further unbounded searches of the project remain. Part II, added in batch 57, shows that deciding radical solvability of an irreducible sextic needs no separating parameter: the sextic is solvable exactly when its triple resolvent has a base-field root at one parameter chosen in advance, or its matching resolvent has one at each of three distinct parameters, because a base-field matching root of a nonsolvable sextic comes from a Galois-invariant pentad, of which there is at most one, and occurs at no more than two parameters.

## 15 Hilbert’s tenth problem

Three reports continuing the formal project `Computability/HilbertTenthProblem`, which proves MRDP in both directions in Lean, the universal pair  $(58, 4)$  and the single-fold unary exponential representation of Jones and Matiyasevich. None of the reports’ own theorems is formalized; each README names the project declarations it builds on.

### 112. Canonical Diophantine certificates

[hilbert-tenth-problem/canonical-diophantine-certificates/article.pdf](#)

from Diophantine\_Causal\_Computation.zip, canonical\_diophantine\_certificates.zip,

ProveIt\_Linear\_Size\_Diophantine\_Certificates.zip, Witness\_Faithful\_Diophantine\_Compilation.zip,

canonical\_trace\_polytopes.zip, diophantine\_counter\_programs.zip (batch 60); batch 61:

Causal\_Diophantine\_Queue\_Compilation.zip; batch 62: Names\_to\_Numbers\_Diophantine\_Heaps.zip,

order\_free\_diophantine.zip, ProveIt\_Priority\_Pumping\_Diophantine.zip,

one\_shared\_fallback\_research.zip; batch 63: Diophantine\_Certificates\_Without\_Histories.zip

Diophantine representations that control their witnesses, not only their existence. For Petri nets and interval-guarded translations, a convex quadratic integer polynomial whose natural zeros are in bijection with the commutation classes of enabled executions of length  $T$  (its real zero set is a rational polytope with exactly those lattice points), with quartic second routes, a Cartier–Foata

variant for bounded causal height, exact commutation tests, #P-completeness of counting and an exponential lower bound for normal-form automata. Single-fold quartics for multisets and nested repetitions of counter actions independent of the repetition counts, for “ $p(i) \geq 0$  for all  $i \leq T$ ” with  $O((d+1)^3)$  witnesses independent of  $T$ , for memory logs with  $24L$  witnesses, and for FRACTRAN and scheduled SKI runs. For cyclic-tag and deletion-tag systems and fixed-read FIFO networks, certificates with one coordinate per consumed symbol and one guard slack, with no intermediate queue quantified; faithful word memory with polynomial insertion at both ends needs two natural coordinates. Proves that a single-fold (finite-fold) universal halting polynomial exists exactly when every enumerable set has such a representation, an open problem not solved here. Parts X–XIII, added in batch 62 from four manuscripts: quartic certificates for dynamic heaps and local graph programs, with birth-order normalization of fresh names, two-stack universality and a proof that perpetual memory safety has no existential representation; for polynomials nonnegative on the real orthant, degrees two and three represent exactly the semilinear sets and degree four every enumerable set, with single-fold quadratics for semilinear sets and order-free self-assembly compilers; for first-enabled-rule multiset and FRACTRAN programs the sharp pumping threshold  $2H + 1$ , a uniform single-fold quartic and c.e.-completeness of eventual periodicity; and conservative priority reaction networks with one shared fallback, with single-fold trace and minimum-fuel quartics and a least-reservoir relation whose single-fold (finite-fold) representability is again equivalent to that of every c.e. set. Part XIV (batch 63): one quartic per run-length routing topology with exactly  $4n + 4m + s$  witnesses and a unique root iff the run terminates, and a fixed polynomial-time rank function whose single-fold (finite-fold) graph is equivalent to the corresponding principle for all c.e. sets. Merged from twelve manuscripts; not a Lean formalization.

### 113. Diophantine laws of probabilistic, quantum and continuous computation

[hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/article.pdf](#)  
 from ProveIt\_Diophantine\_Quantum\_Normalization.zip, quantum\_diophantine\_research.zip,  
 ProveIt\_Diophantine\_Barriers.zip (batch 60); batch 62: Quartic\_Probability\_Landscapes.zip,  
 ProveIt\_Reversible\_Equilibria.zip, Diophantine\_Certificates\_for\_Coherent\_Computation.zip,  
 proveit\_rational\_rotations\_research.zip, dense\_rotations\_research.zip,  
 strict\_contractions\_research.zip, Diophantine\_Contracting\_2Adic.zip,  
 Precision\_Is\_Memory\_Diophantine\_Neural\_Computation.zip; batch 64:  
 Rational\_Tubes\_Integer\_Zeros.zip

One fixed quartic represents the output law of every effective probabilistic or quantum program as masses of projected roots; exact normalization budgets with a unique-witness quartic certificate; normalized two-outcome probabilities are exactly the weakly computable reals, and conditional comparisons are  $\Sigma_2^0$ -complete. Deciding whether a seven-outcome rational quantum instrument in dimension four has an impossible outcome word is undecidable (via Neary’s matrix-mortality theorem), with bijective quartics for words of fixed length. One rational piecewise-affine map of the cube realizes every disjoint pair of enumerable sets as exact acceptance and robust rejection, with quartic safety certificates for a computable critical-noise radius. Parts V–VII, added in batch 62 from eight manuscripts: Boolean circuits compile to coercive integer quartics of height at most 4 whose real zeros are Boolean, with uniform Hessian and rounding constants, so that halting probabilities of fair-coin programs are root counts, component counts or Euler characteristics; birth–death chains with defect environments have the sharp spectral gap  $(3 - 2\sqrt{2})/4$  and a root mass that is rational exactly for eventually periodic environments; unique-witness quartics for coherent contraction trees over  $\mathbb{Z}[e^{i\pi/4}]$ , and dense rational  $\text{SO}(4)$  gate groups from the rotations  $(3 \pm 4i)/5$ ,  $(3 \pm 4j)/5$  with c.e.-complete state transfer or c.e.-complete membership; a fixed  $1/2$ -Lipschitz rational piecewise-linear map of  $\mathbb{R}^4$  and a  $1/2$ -contraction of  $\mathbb{Z}_2$  with  $\Sigma_1^0$ -complete exact half-space reachability and exact extinction; and unique-witness quadratics for rational clipped neural networks, whose tied four-layer synthesis is equivalent to Hilbert’s tenth problem over  $\mathbb{Q}$ . Part VIII (batch 64): complete integer tube certificates, sums of squares of quadratics with at most one witness, for reaching open rational targets under rational polynomial flows, and global mass-action normal forms that carry them to chemical reachability. Merged from twelve manuscripts; not a Lean formalization.

#### 114. Beyond halting: recurrence, progress deadlines and Diophantine liveness

[hilbert-tenth-problem/liveness-beyond-halting/article.pdf](#)

from Beyond\_Halting\_Diophantine\_Liveness.zip, Quadratic\_Diophantine\_Dynamics.zip (batch 62);  
batch 63: quadratic\_infinite\_computation.zip

For computable finitely branching systems, existence of a recurrent run with a computable progress deadline is  $\Sigma_3^0$ -complete and equivalent to a nonempty effectively closed class of accepting schedules, and to an accepting schedule of hyperimmune-free degree; fixed deadlines are  $\Pi_1^0$ , bounded gaps  $\Sigma_2^0$ , unrestricted recurrence  $\Sigma_1^1$ -complete, with strict separations and every c.e. degree as an exact deadline spectrum. An auxiliary-free degree-two polynomial with coefficients at most eight gives the transition relation of counter networks (degree two is minimal); a second, incoming-edge encoding has height six, a sum-of-squares quartic companion and an exact planar realization; and a two-stack quartic certifies finite runs meeting prescribed deadlines. Merged from three manuscripts; not a Lean formalization.